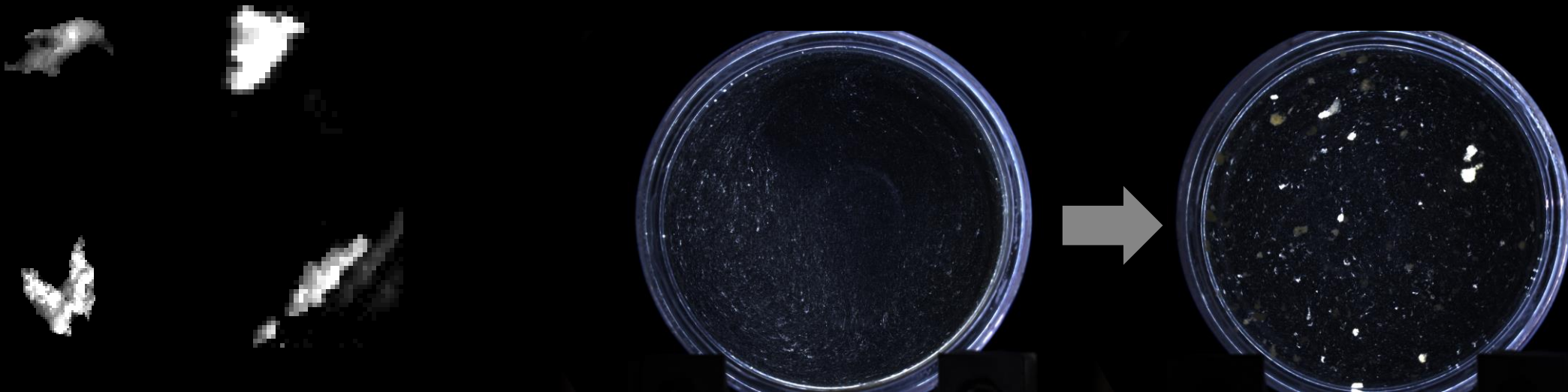
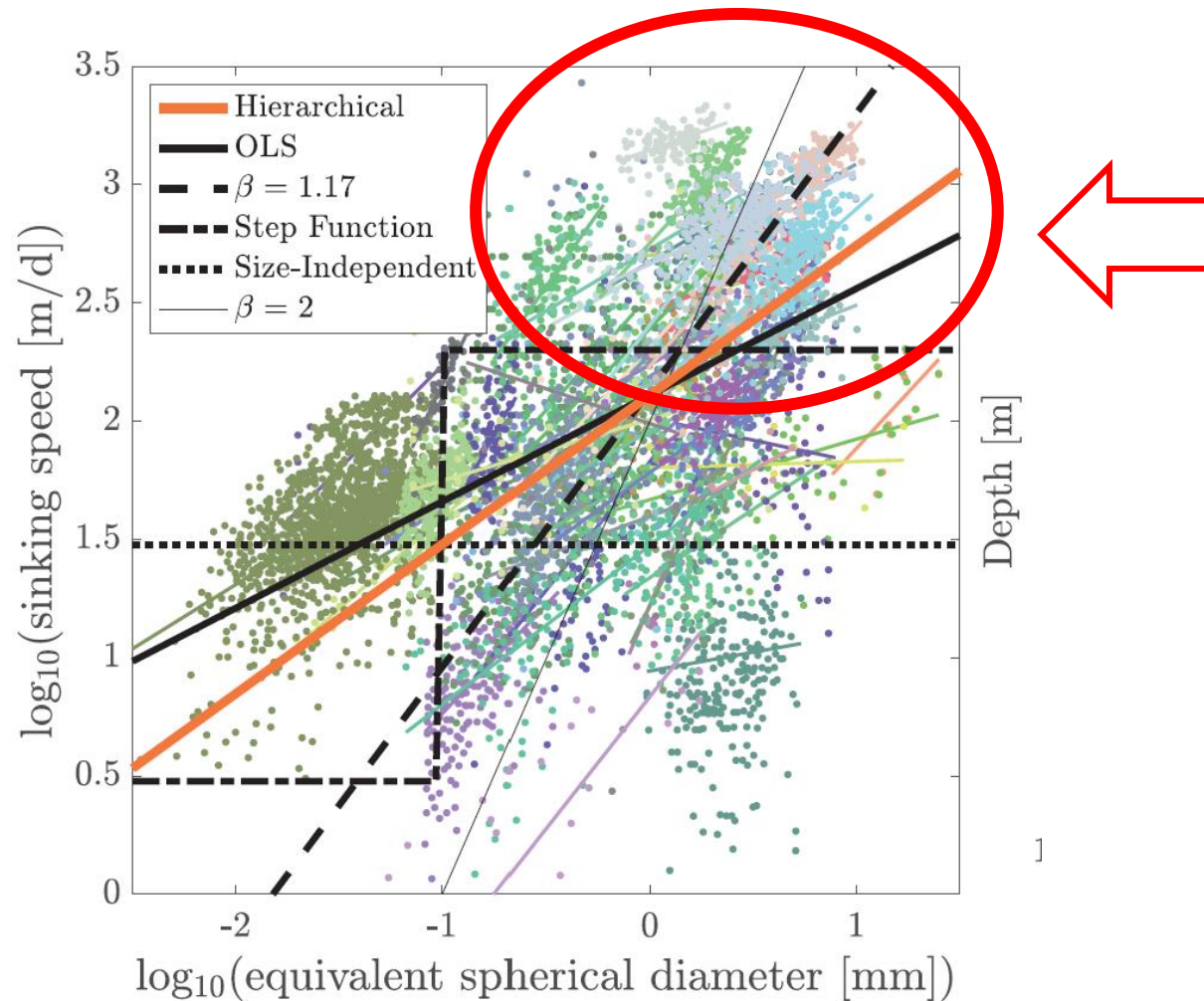


The role of aggregation and disaggregation in carbon transport

Matthew Rau, *The George Washington University*
(matthew.rau@gwu.edu)



Big particle aggregates can settle very quickly (and also slowly...)



- Large particles can settle 100s to 1000s of meters per day
- Many variables go into determining sinking speed, but size is still very important
- Aggregation is the primary way that small particles are incorporated into big particles, i.e. aggregates

Cael et. al, (2021)

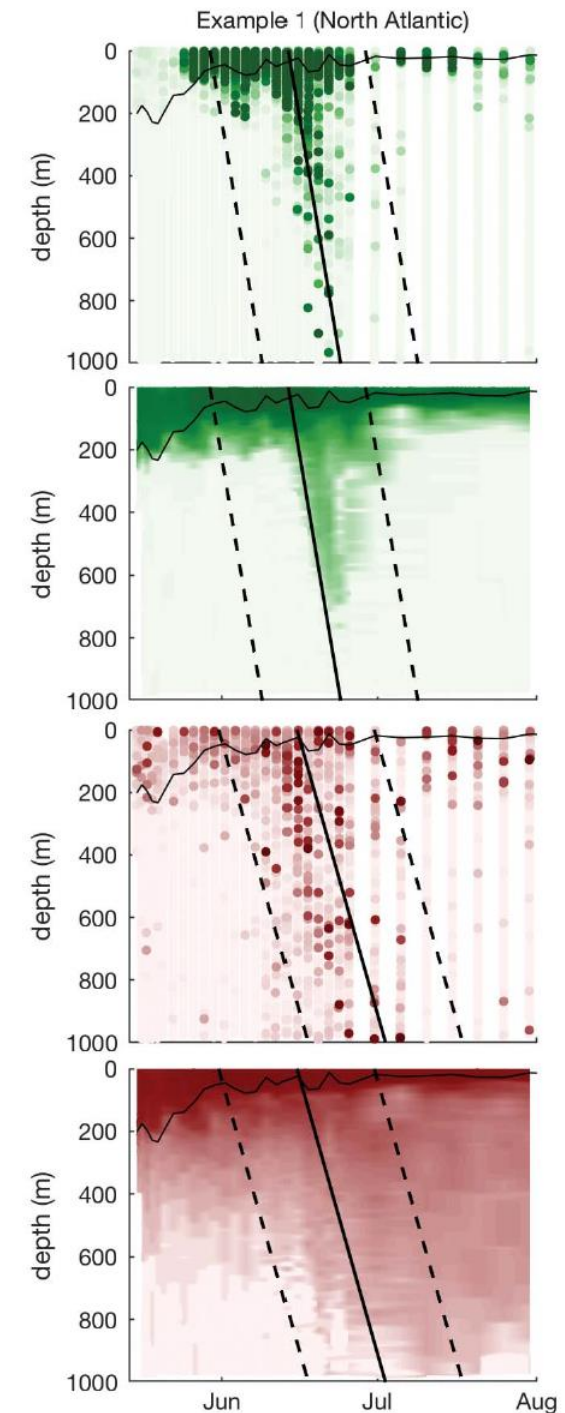
Aggregation is super prevalent in the ocean

RESEARCH

CARBON CYCLE

Major role of particle fragmentation in regulating biological sequestration of CO₂ by the oceans

Nathan Briggs^{1,2*}, Giorgio Dall'Olmo^{3,4}, Hervé Claustre²



Aggregation is super prevalent in the ocean



Global Biogeochemical Cycles*

RESEARCH ARTICLE
10.1029/2025GB008676

Key Points:

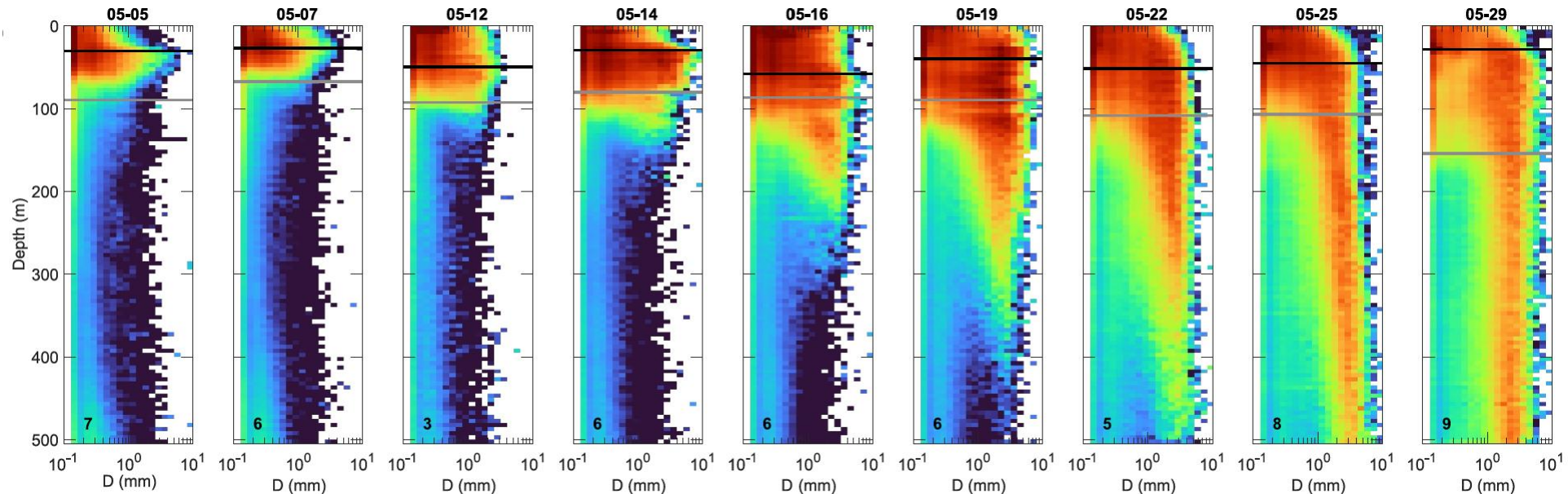
- Marine snow is highly ephemeral in the ocean's mixed layer, responding rapidly to changes in turbulence due to the passage of storms

Assessing Marine Snow Dynamics During the Demise of the North Atlantic Spring Bloom Using In Situ Particle Imagery

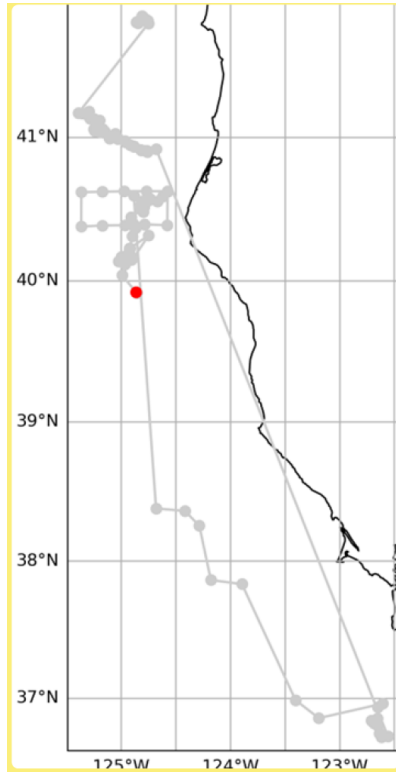
D. A. Siegel¹ , A. B. Burd² , M. L. Estapa³, E. Fields¹ , L. Johnson⁴, U. Passow⁵ , E. Romanelli^{1,6} , M. A. Brzezinski⁷ , K. O. Buesseler⁸ , S. J. Clevenger⁹, I. Cetinic^{10,11}, L. Drago^{12,13} , C. A. Durkin¹⁴ , R. Kiko^{15,16} , S. J. Kramer^{14,17}, A. E. Maas¹⁸, M. M. Omand¹⁹ , and D. K. Steinberg²⁰ 

Storms on:

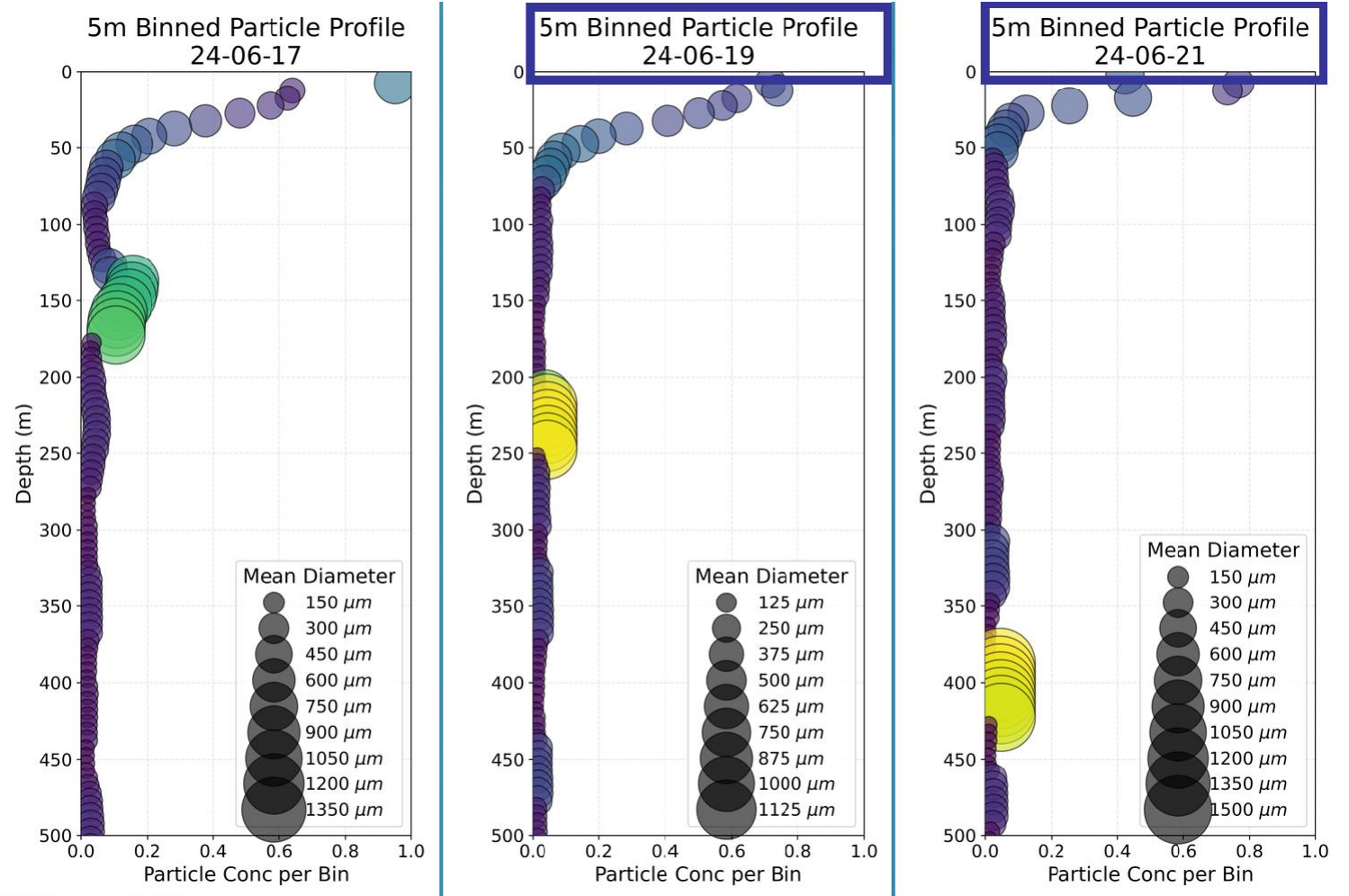
- 05-09/10
- 05-15
- 05-20
- 05-23



Aggregation is super prevalent in the ocean



Holography particle size profiles
from VIISTA1 cruise
CA current, June 2024



Aggregation rates are driven largely by particle and fluid flow properties

The Population Balance Equation (PBE)

$$\frac{dN(d_p, t)}{dt} = \frac{\alpha}{2} \int_0^{d_p} \beta(d'_p, d_p - d'_p, t) N(d'_p, t) N(d_p - d'_p) dd'_p$$

Aggregation

$$- \alpha \int_0^{\infty} \beta(d_p, d'_p) N(d'_p, t) N(d_p, t) dd'_p$$

Breakup

$$+ \int_{d_p}^{\infty} K(d'_p) \Gamma(d_p, d'_p) N(d'_p) dd'_p$$

$$- K(d_p) N(d_p, t)$$

Settling

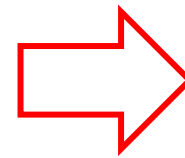
$$- N(d_p, t) \frac{w_s(d_p)}{Z}$$

Particle Aggregation

Adrian B. Burd¹ and George A. Jackson²

¹Department of Marine Sciences, University of Georgia, Athens, Georgia 30602-3636;
email: adrianb@uga.edu

²Department of Oceanography, Texas A&M University, College Station, Texas 77843-3146;
email: gjackson@tamu.edu



$$Agg. Rate = f(\alpha, \beta, N)$$

Aggregation needs lots of colliding sticky particles

N – [# / L], particle number concentration in a specific size class

- Transparent Exopolymer Particles (TEP) can contribute to these #s

α – ‘stickiness’, a dimensionless quantity heavily dependent on particle population

- inherent ‘particle’ surface properties like hydrophobicity
- EPS quantity and content
- phytoplankton health
- other microorganisms and remineralization

β – [1/s], aggregation kernel, or the rate at which particles contact one another

- Governed by hydrodynamics and particle size

Marine biopolymer self-assembly: implications for carbon cycling in the ocean†

Pedro Verdugo,^{*a} Monica V. Orellana,^b Wei-Chun Chin,^c Timothy W. Petersen,^b Ger van den Eng,^d Ronald Benner^c and John I. Hedges^d

Received 4th January 2008, Accepted 9th January 2008

First published as an Advance Article on the web 23rd April 2008

DOI: 10.1039/b800149a

Particle Aggregation

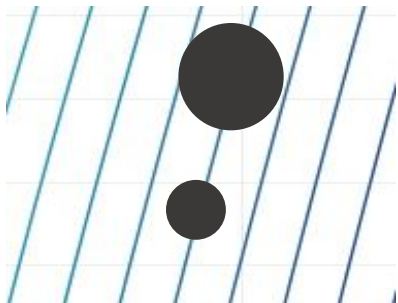
Adrian B. Burd¹ and George A. Jackson²

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email: adrianb@uga.edu

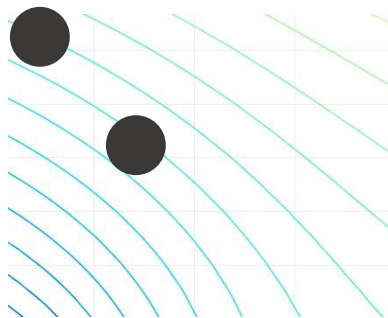
²Department of Oceanography, Texas A&M University, College Station, Texas 77843-3146;
email: gjackson@tamu.edu

Hydrodynamics drives the rate of particle contact

Differential sedimentation



Laminar or turbulent shear



Particle Aggregation

Adrian B. Burd¹ and George A. Jackson²

¹Department of Marine Sciences, University of Georgia, Athens, Georgia 30602-3636;
email: adrianb@uga.edu

²Department of Oceanography, Texas A&M University, College Station, Texas 77843-3146;
email: gjackson@tamu.edu

Table 1 Expressions for the coagulation kernels as functions of the particle radius (r). In these formulae, k is Boltzman’s constant, T is the absolute temperature of the water, μ is the dynamic viscosity, γ is the shear gradient, ν is the kinematic viscosity, ε is the turbulent energy dissipation rate, and w_i is the settling velocity of a particle with radius r_i . Note that the fractal relationship (Equation 8) should be used to convert between the mass and radius for a given particle

Mechanism	Rectilinear kernel	Curvilinear kernel
Brownian motion	$\frac{2}{3} \frac{kT}{\mu} \frac{(r_i+r_j)^2}{r_i r_j}$	–
Laminar shear	$\frac{4}{3} \gamma (r_i + r_j)^3$	
Turbulent shear	$1.3 \left(\frac{\varepsilon}{\nu}\right)^{1/2} (r_i + r_j)^3$	$9.8 \frac{p^2}{1+2p^2} \left(\frac{\varepsilon}{\nu}\right)^{1/2} (r_i + r_j)^3$
Differential sedimentation	$\pi (r_i + r_j)^2 w_j - w_i $	$0.5 \pi r_i^2 w_i - w_j $

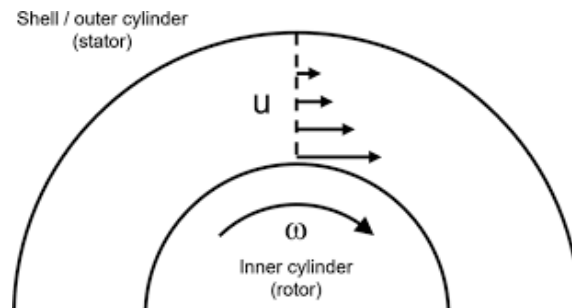
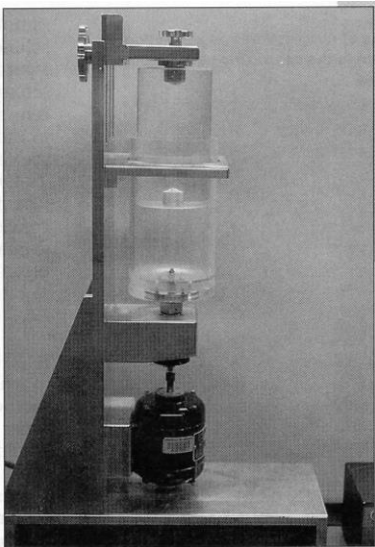
Aggregation kernels are highly idealized and assume particles are smaller than turbulence eddies in the ocean

Most relevant β kernels for oceanography:

Laminar shear

γ determined from velocity gradient

$$\beta = \frac{4}{3} \gamma (r_i + r_j)^3$$



Turbulent shear

Camp-Stein eq: $\gamma \sim \left(\frac{\varepsilon}{\nu}\right)^{1/2}$

$$\beta = 1.3 \left(\frac{\varepsilon}{\nu}\right)^{1/2} (r_i + r_j)^3$$

Kolmogorov lengthscale η
(i.e. smallest eddies)

$$\eta = \left(\frac{\nu^3}{\varepsilon}\right)^{1/4}$$

LIMNOLOGY and OCEANOGRAPHY

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Limnol. Oceanogr. 67, 2022, 348–363
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doi: 10.1002/lno.11996

Oceanic turbulence from a planktonic perspective

Peter J. S. Franks ,* Bryce G. Inman , Jennifer A. MacKinnon, Matthew H. Alford, Amy F. Waterhouse
Scripps Institution of Oceanography, University of California San Diego, La Jolla, California

Mixed layer: $10^{-8} < \varepsilon < 10^{-6}$ W/kg, $3 < \eta < 1$ mm

Below ML: $10^{-10} < \varepsilon < 10^{-8}$ W/kg, $10 < \eta < 3$ mm

Different lab devices have been used to study aggregation: e.g. Jar tests, Taylor-Couette cylinders, and roller tanks

Jar testing for water treatment protocols



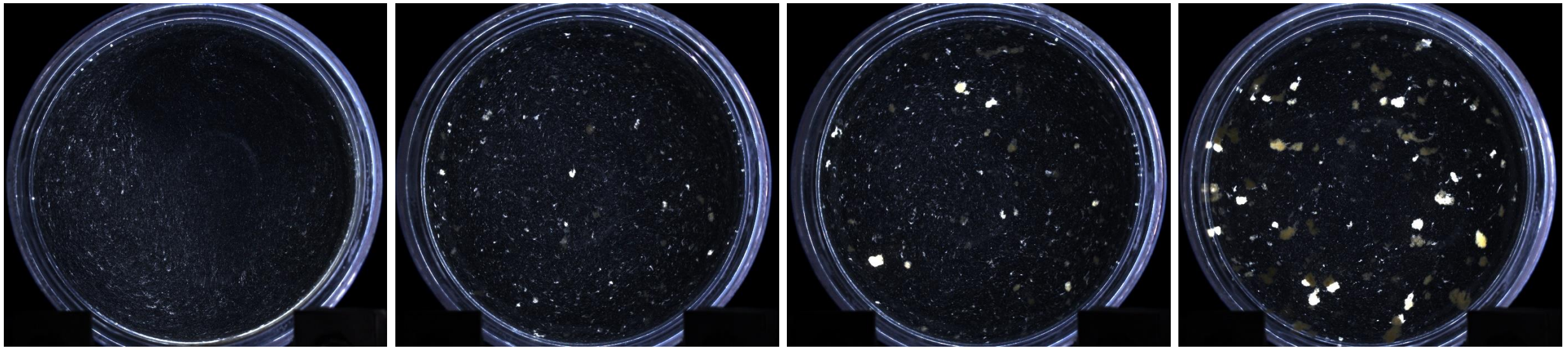
Roller tanks are used widely in marine science to form ‘marine snow’

GLOBAL BIOGEOCHEMICAL CYCLES, VOL. 20, GB1013, doi:10.1029/2005GB002579, 2006

Accumulation of mineral ballast on organic aggregates

Uta Passow¹ and Christina L. De La Rocha¹

Received 24 June 2005; revised 11 October 2005; accepted 4 January 2006; published 28 February 2006.



4 hrs of rolling CTD water from the CA current, June 2024

M. Rau, MOB 2026, SNO Workshop

Fragmentation of marine snow is complex and poorly understood

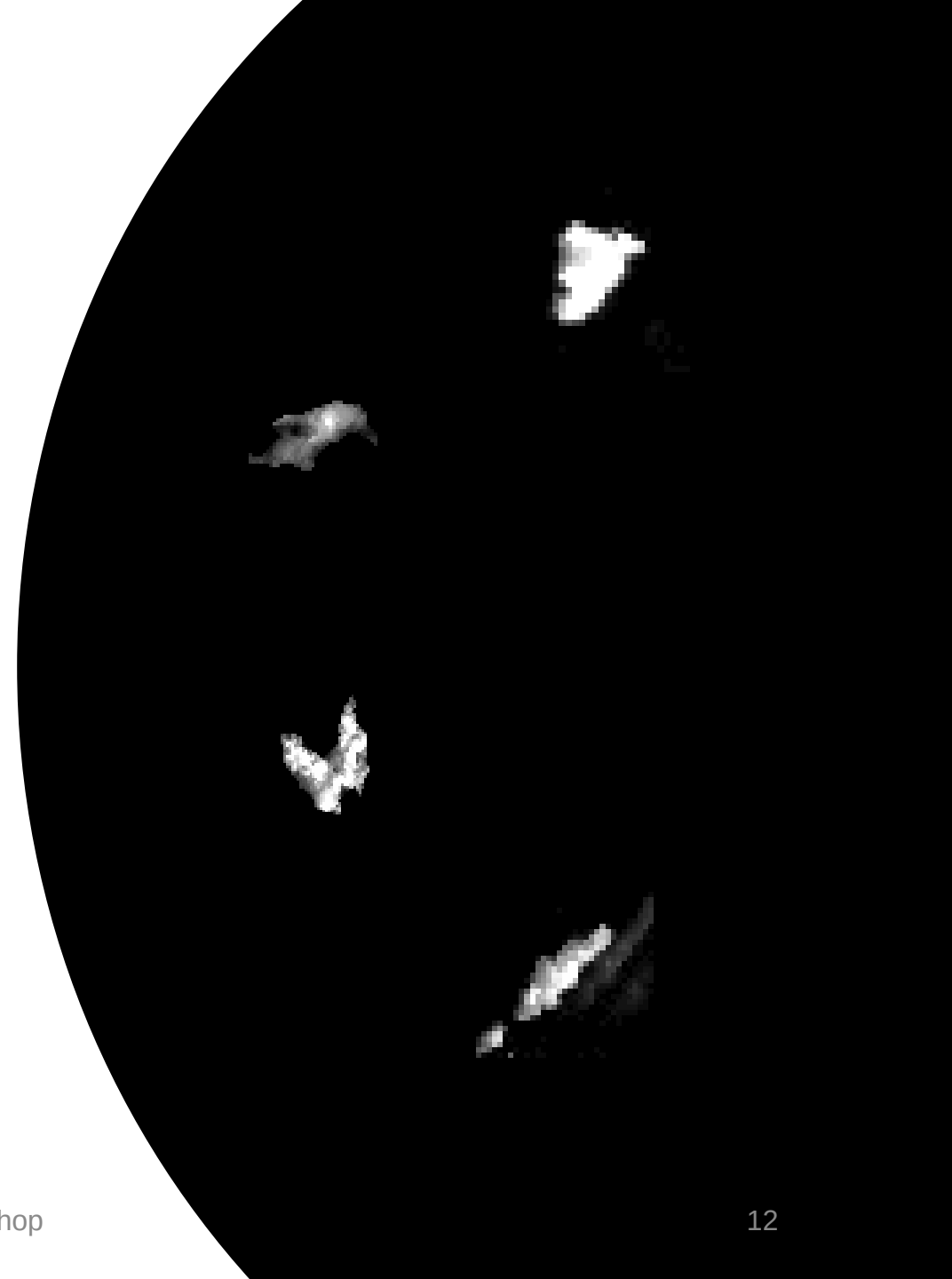
RESEARCH

CARBON CYCLE

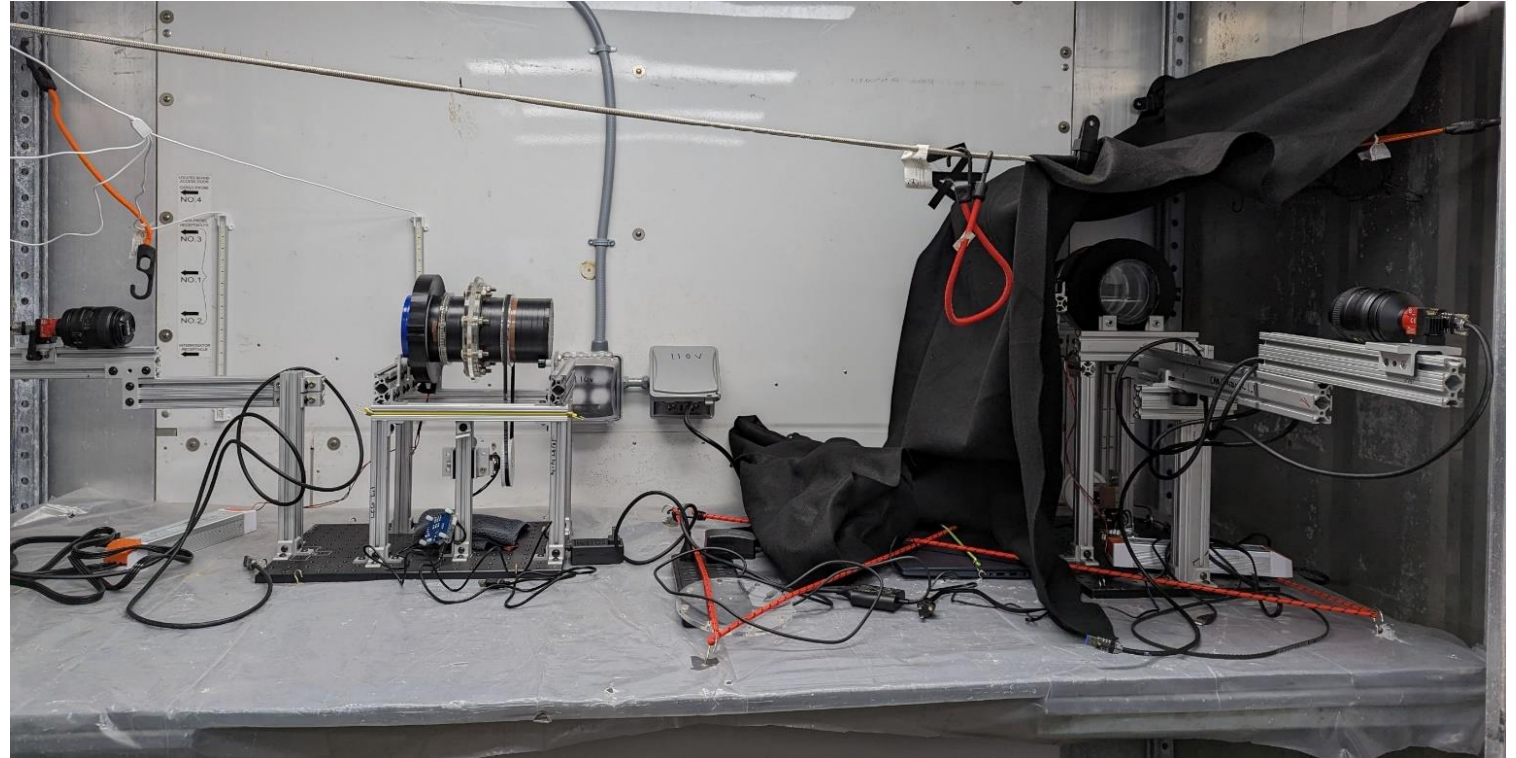
Major role of particle fragmentation in regulating biological sequestration of CO₂ by the oceans

Nathan Briggs^{1,2*}, Giorgio Dall'Olmo^{3,4}, Hervé Claustre²

- Fragmentation plays a role in flux attenuation, but the mechanisms driving fragmentation is unclear
- Turbulence shear is small, but episodic events like internal waves and zooplankton feeding can fragment aggregates
- Aggregates are viscoelastic – they deform and ‘creep’ continually, changing morphology
- The change in strength as aggregates degrade and remineralize is unclear



Roller tanks are ubiquitous, but the details of their usage matters



The flow in a roller tank approximates 'solid body rotation' but this is a simplification



Pergamon

Deep-Sea Research I, Vol. 41, No. 3, pp. 429-437, 1994
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0967-0637/94 \$7.00 + 0.00

Particle trajectories in a rotating cylinder: implications for aggregation incubations

GEORGE A. JACKSON*

(Received 8 February 1993; in revised form 16 June 1993; accepted 16 July 1993)

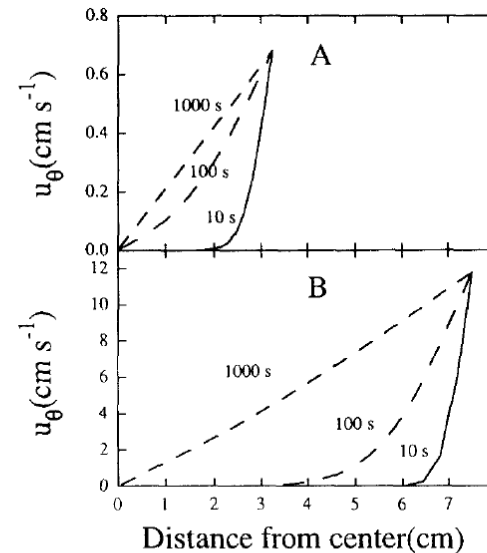
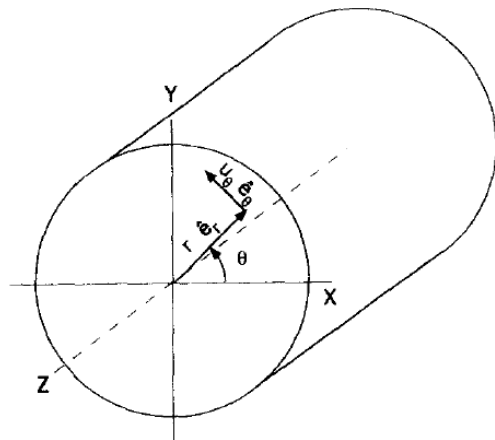


Fig. 2. Fluid velocity as a function of relative distance from the cylinder axis at different times. The values shown are: Case 1 (A) $a = 7.5$ cm, $\omega = 1.57$ s⁻¹, and $\nu = 0.01$ cm² s⁻¹; Case 2 (B) $a = 3.25$ cm, $\omega = 0.21$ s⁻¹. Case 1 values typical of the systems used by SHANK and EDMONDSON (1989); Case 2 values typical of more recent systems considered for use (PASSOW and WASSMAN, personal communication). u_θ for steady state (solid body rotation) would be a straight line.

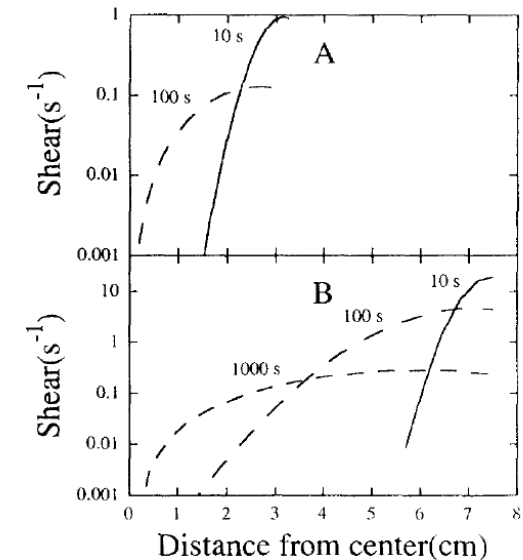


Fig. 3. Fluid shear as a function of relative distance from the cylinder axis for different times. Parameters are the same as those shown in Fig. 2.

The flow in a roller tank approximates 'solid body rotation' but this is a simplification

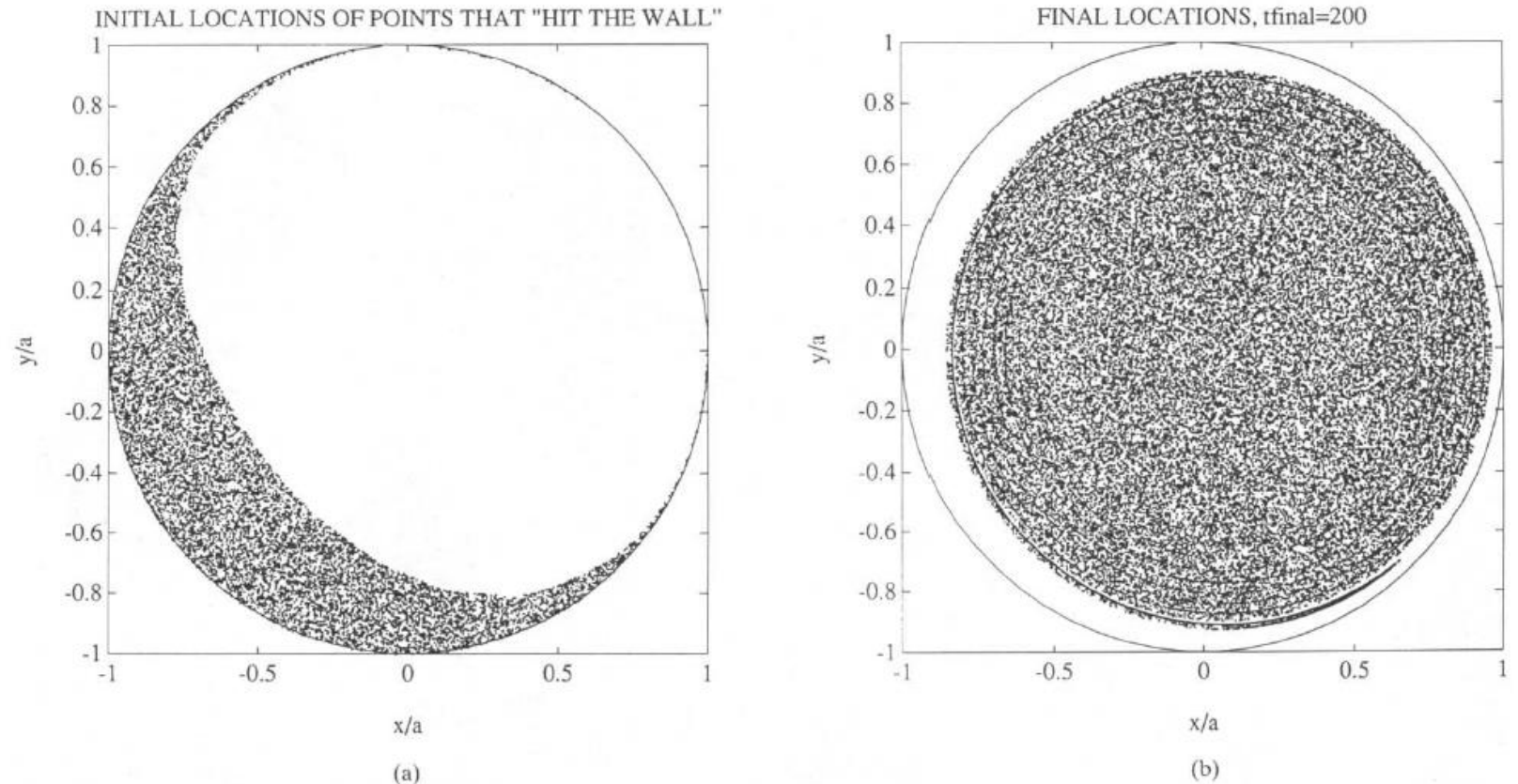
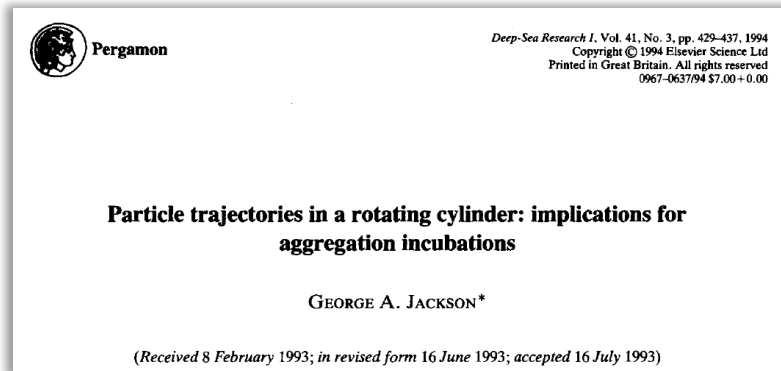


Fig. 6. Distributions from 500,000 particles initially distributed randomly throughout the cylinder before spin up. (A) initial location of particles which hit the cylinder; (B) location of all particles after the initial transients have dissipated. Notice that there is a spiral of particles in the outer part of the final mass of particles that is formed by the particles that hit the cylinder wall. Parameters as in Fig. 5.

The flow and confined nature of the roller tank means it is not a good simulation of the real ocean, but still a useful tool

**LIMNOLOGY
and
OCEANOGRAPHY: METHODS**

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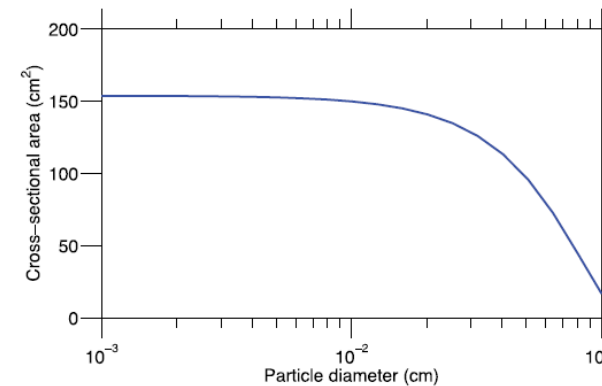
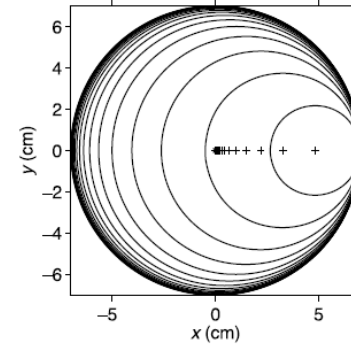
Limnol. Oceanogr.: Methods 13, 2015, 194–201
© 2015 The Authors Limnology and Oceanography: Methods
published by Wiley Periodicals, Inc. on behalf of Association for the
Sciences of Limnology and Oceanography
doi: 10.1002/lom3.10018

Coagulation in a rotating cylinder

George A. Jackson*

Department of Oceanography, Texas A&M University, College Station, Texas

“Coagulation rates are initially slower in the rotating system because of the lack of shear, but rates increase as coagulation creates larger particles that become concentrated within smaller regions of the rotating vessel. [...] In addition, coagulation for larger particles within the rotating tanks is faster than in nature.”



Stable region size depends on particle settling velocity (i.e. size)

Fig. 3. The cross-sectional areas associated with different particle trajectories in a rotating tank reactor. Top: particle trajectories; bottom: cross sectional area as a function of apparent particle diameter. The stable points in the trajectories were chosen to correspond with the midpoints of the sections used in the coagulation simulations; successive points are for particles with twice the mass of those for the preceding points. The settling velocities calculated in the coagulation simulations. $\Omega = 0.31 \text{ rad s}^{-1}$ (3 RPM); $r_o = 7 \text{ cm}$.

You can even extract settling velocities from roller tanks

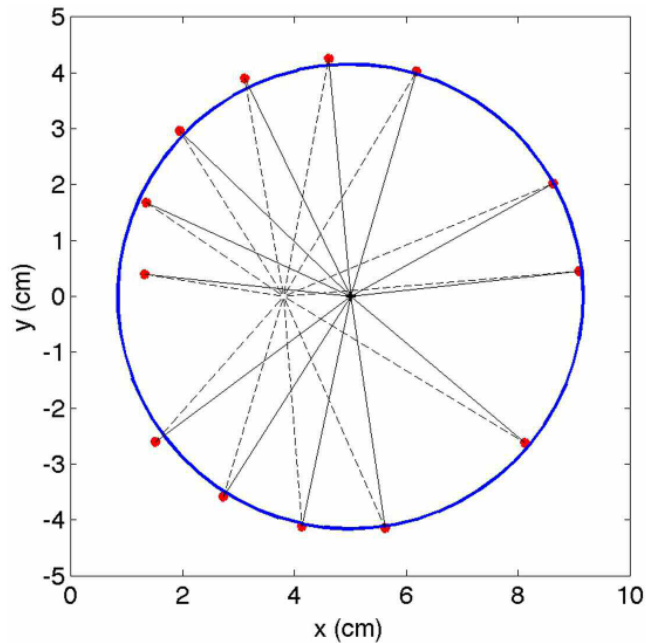


Fig. 1. Example of orbit analysis to derive sinking velocity of particles in roller tanks. The distances between the observed aggregate positions along its orbit (red dots) and a chosen orbit center (at $x = 3.8$ cm, $y = 0$ cm; linked to observations by broken lines) are squared and summed up. The position of the orbit center is varied on a grid with fine (0.1 mm) resolution and the corresponding sum of the squared distances is calculated. The point where this sum is minimal is used as the optimal orbit center position (at $x = 5$ cm, $y = 0$ cm; linked to observations by solid lines). The mean distance between the optimal center and the observations is the optimal radius (4 cm) used to plot the circle (blue solid line) around the orbit center. The distance between the axis of the roller tank at (0.0) and the aggregate's center position is 5 cm. With a rotation rate of 0.083 s^{-1} (5 rpm), the aggregate's sinking velocity is $5 \text{ cm} \times 0.083 \text{ s}^{-1} = 0.41 \text{ cm s}^{-1}$ or 180 m d^{-1} .

LIMNOLOGY and OCEANOGRAPHY: METHODS

Limnol. Oceanogr.: Methods 8, 2010, 386–393
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A novel method to measure particle sinking velocity in vitro, and its comparison to three other in vitro methods

Helle Ploug^{1,2}, Anja Terbrüggen¹, Anna Kaufmann¹, Dieter Wolf-Gladrow¹, and Uta Passow^{1,3*}

¹Alfred Wegener Institute for Polar and Marine Research, Am Handelshafen 12, D - 27570 Bremerhaven, Germany

²Stockholm University, Dept of Botany, Lilla Frescativägen 5, 10691 Stockholm

³Marine Science Institute, University of California Santa Barbara, CA 93106, USA

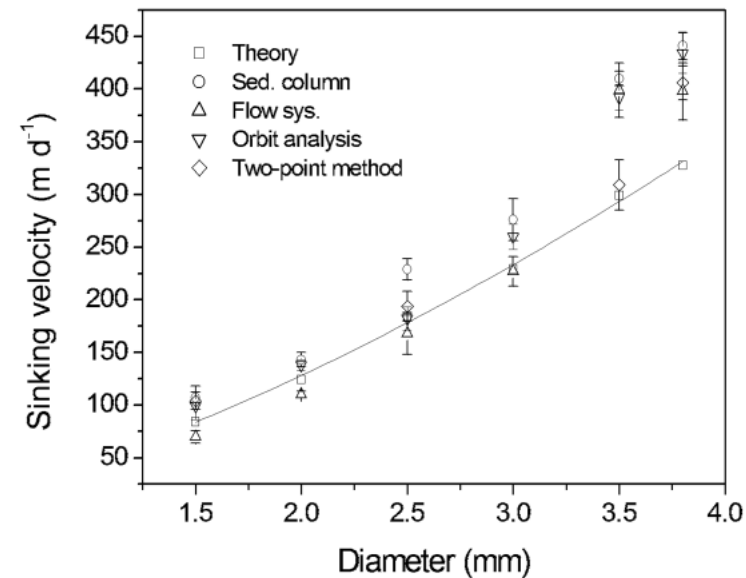
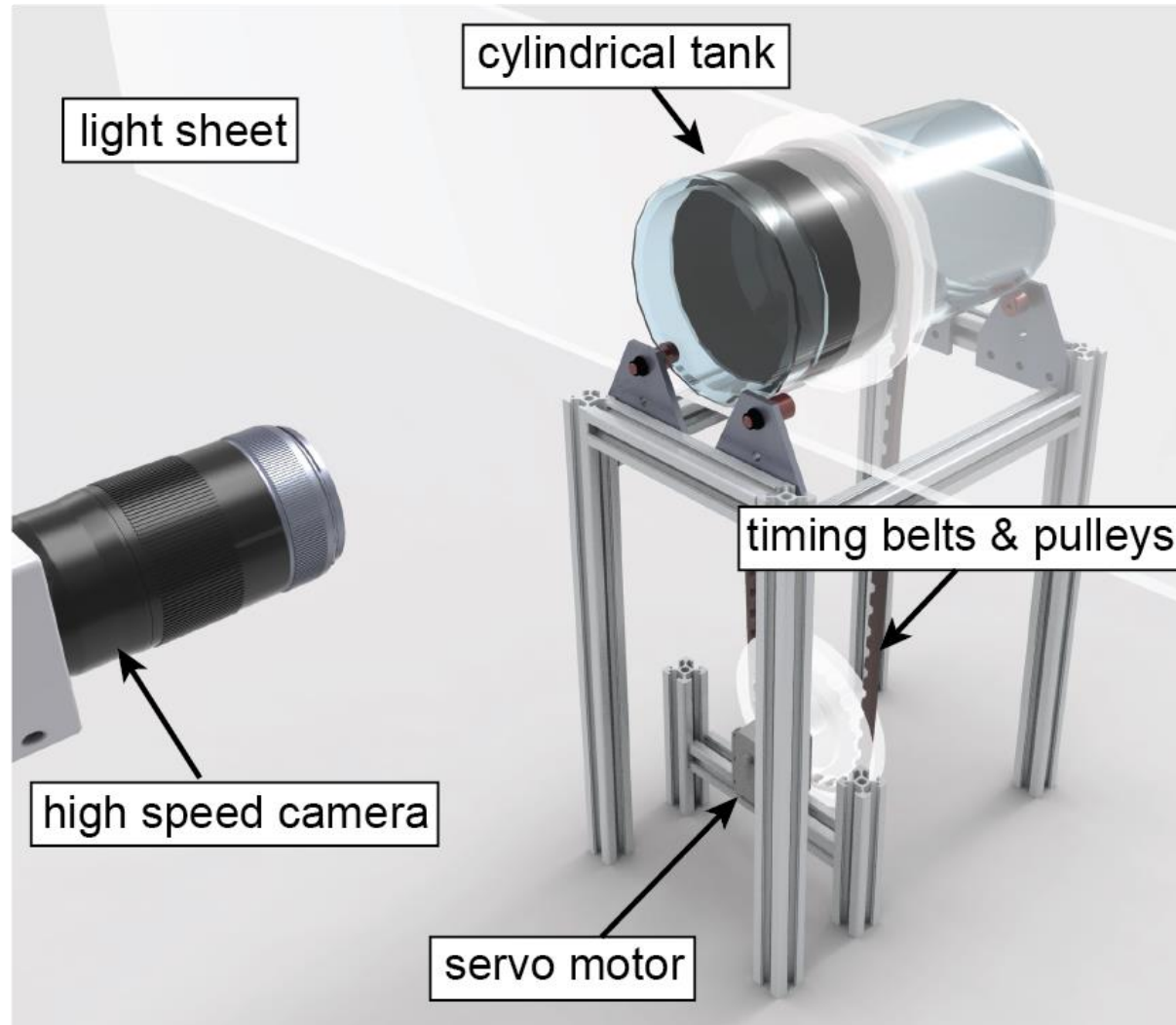


Fig. 2. Sinking velocity of agar spheres as a function of sphere diameter.

Which means you need good imagery with calibrated depths to get quantifiable data



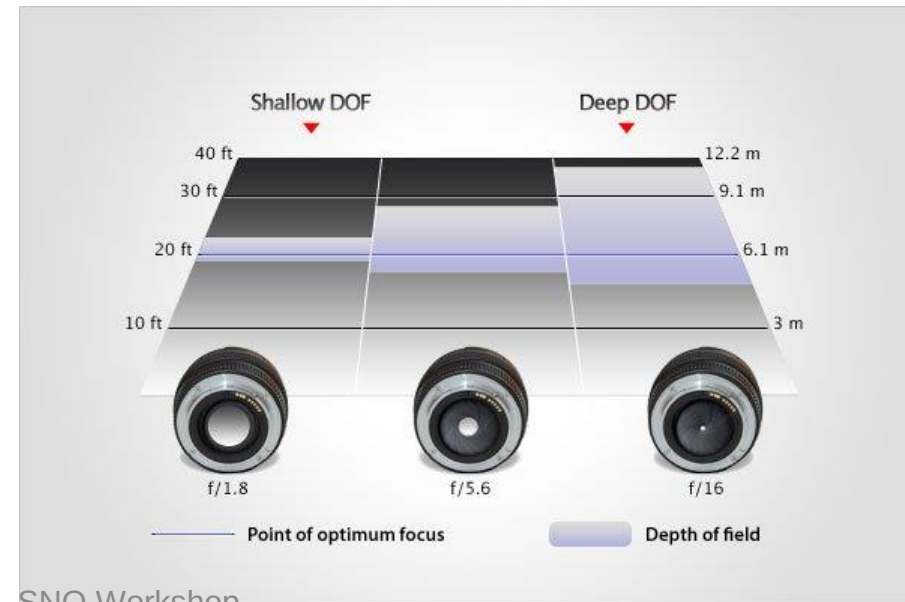
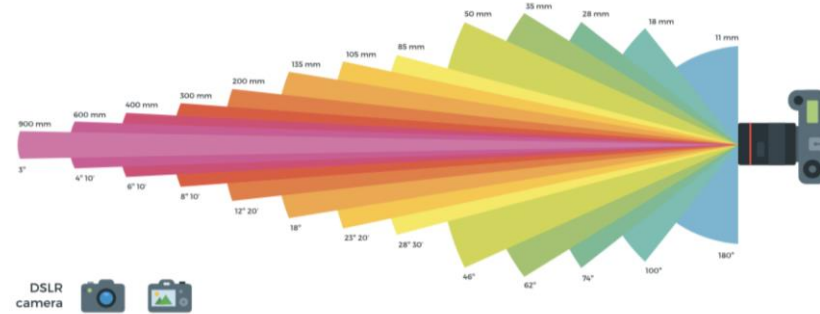
Note on imaging

Most particle tracking with a single camera assumes planar projections and 2D motion.

Best to use:

- High focal length lenses
- Wide open apertures (lenses with manual apertures)
- 2D illumination

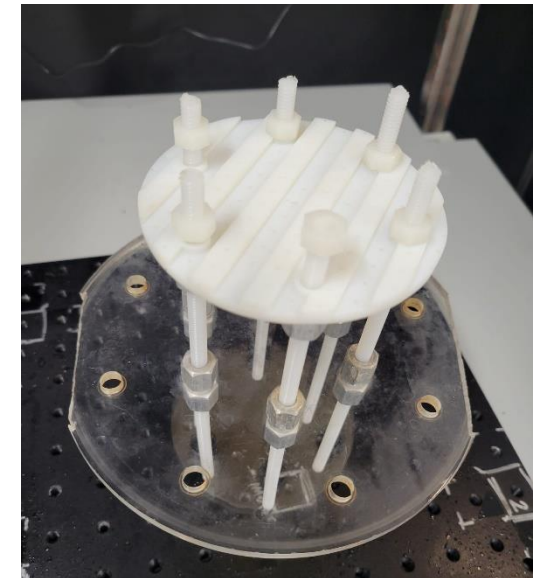
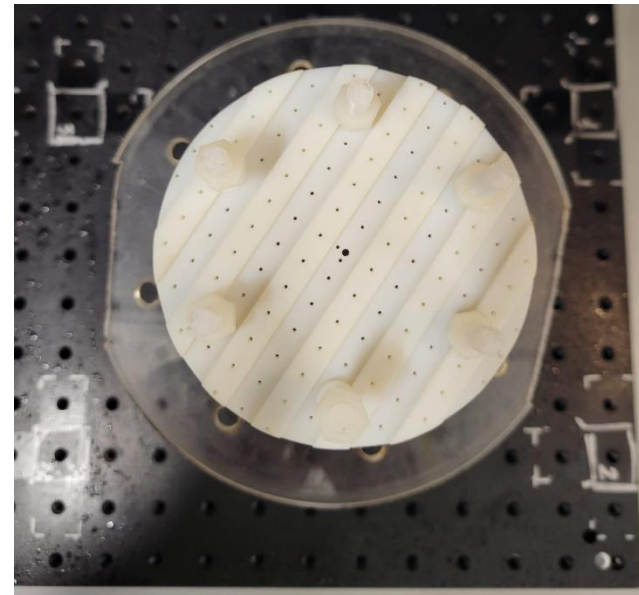
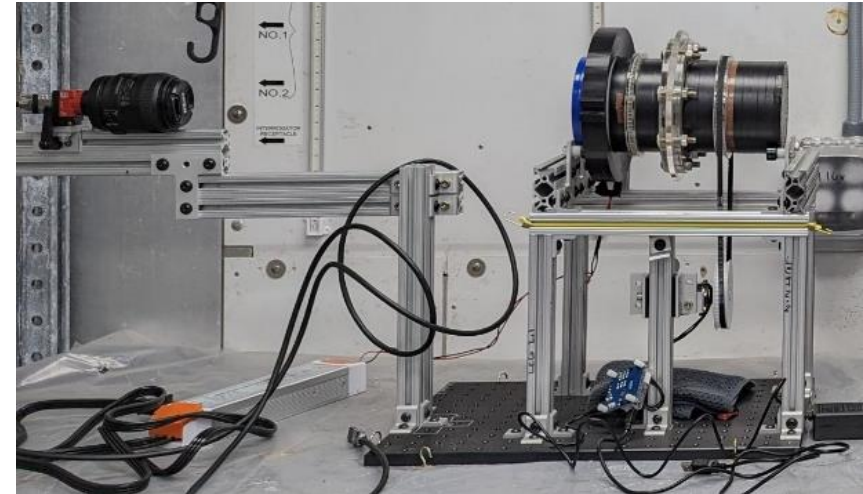
Lens Focal Length
in mm and angle of view in degrees



Note on calibrating

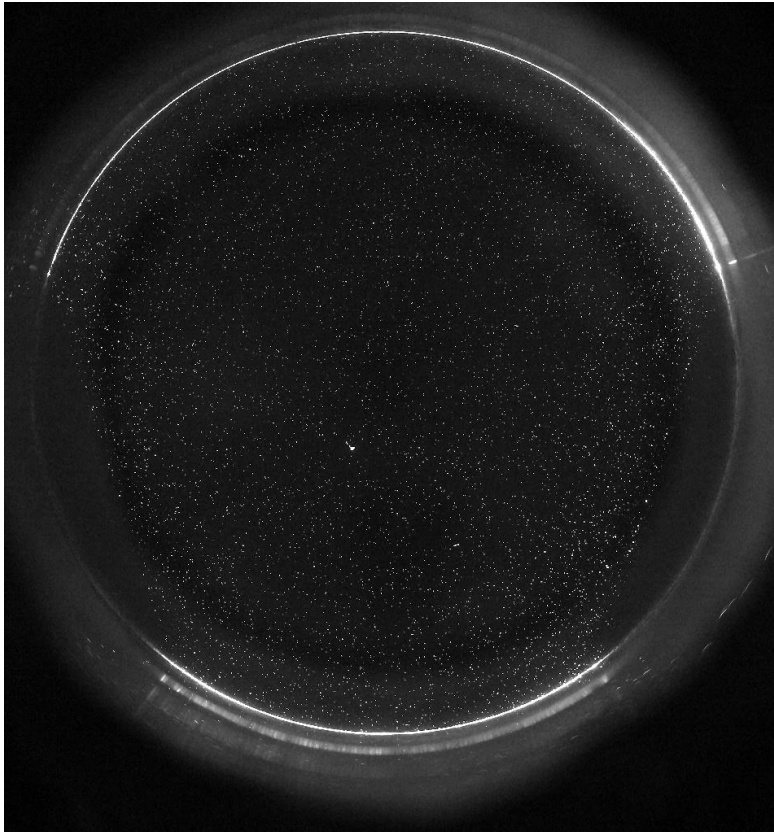
You must know your pixel-to-length magnification at the plane you illuminate

- Must calibrate with water in the tank due to distortion

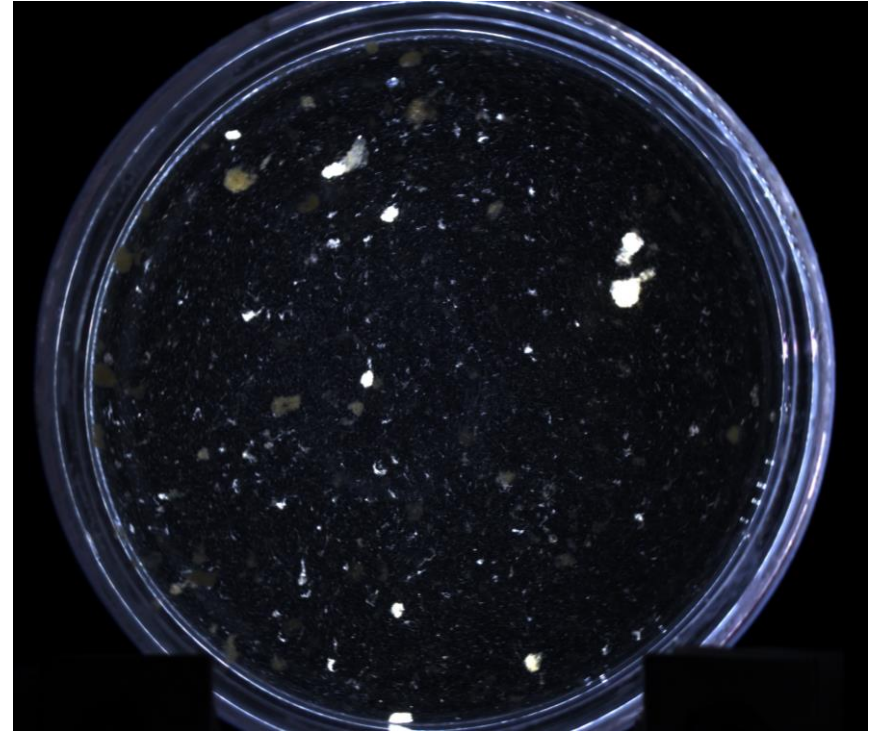


Laser sheets are great for 2D illumination, but they are usually too thin to see large aggregates

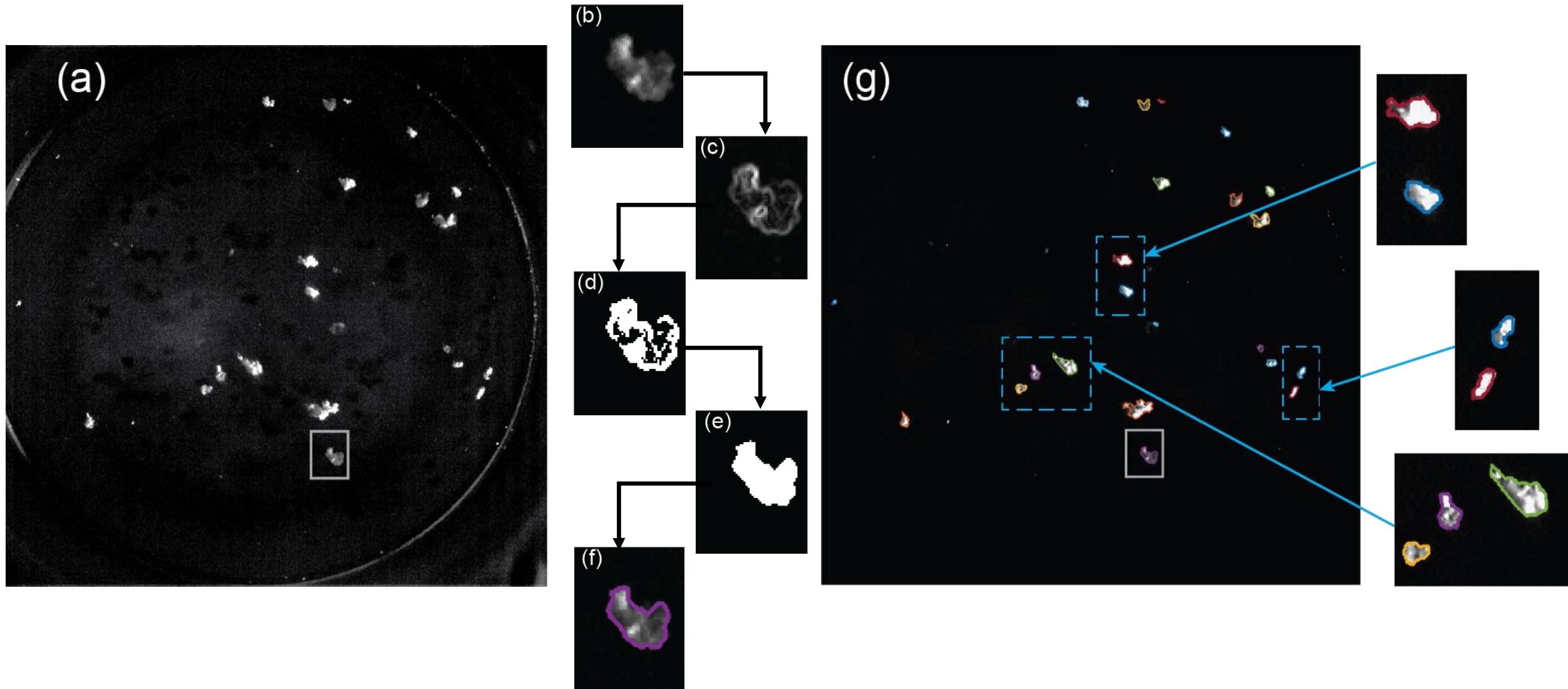
Laser sheet



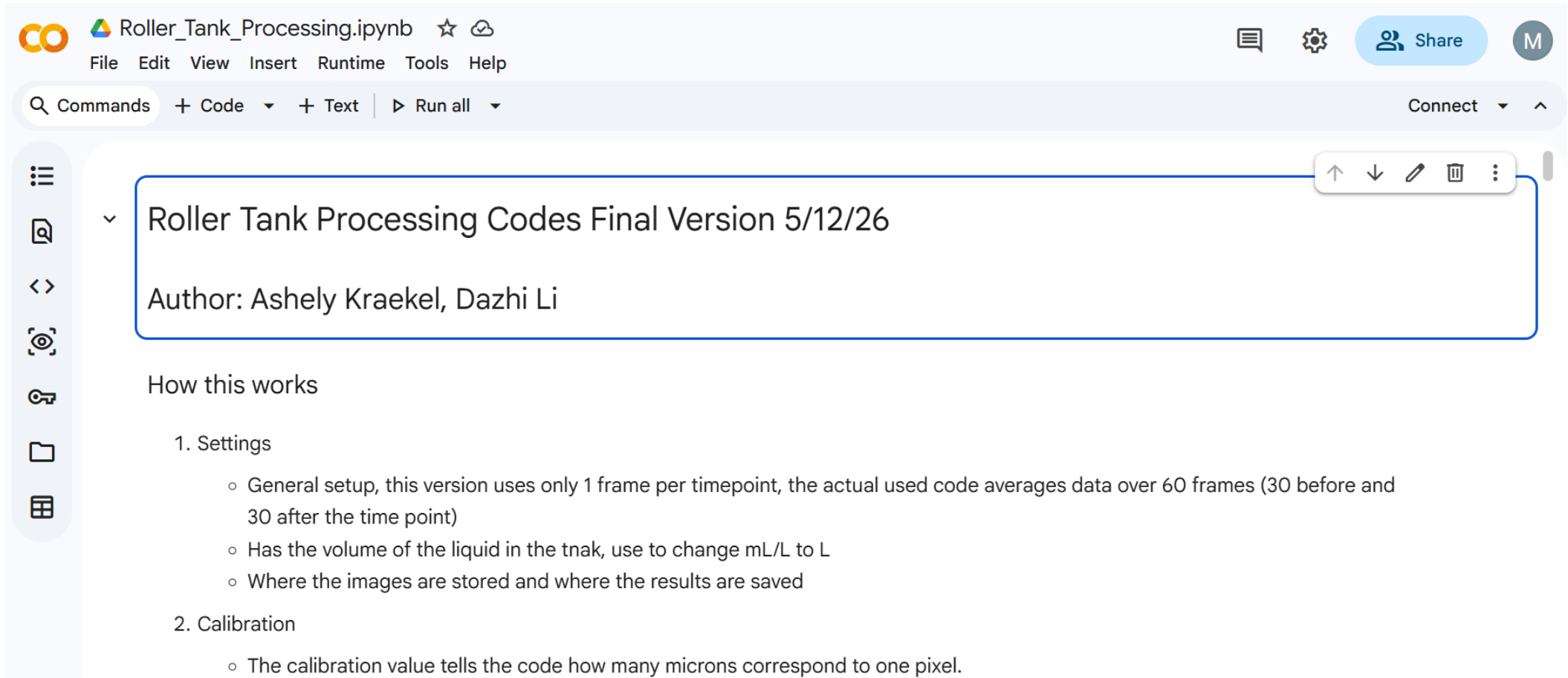
Ring LED light



Traditional image processing steps can be used to quantify particle size – pay attention to your lower particle resolution!



We use an algorithm in Python to quantify PSD and total aggregate volume changes with time



Roller_Tank_Processing.ipynb

File Edit View Insert Runtime Tools Help

Q Commands + Code + Text | ▶ Run all

Connect

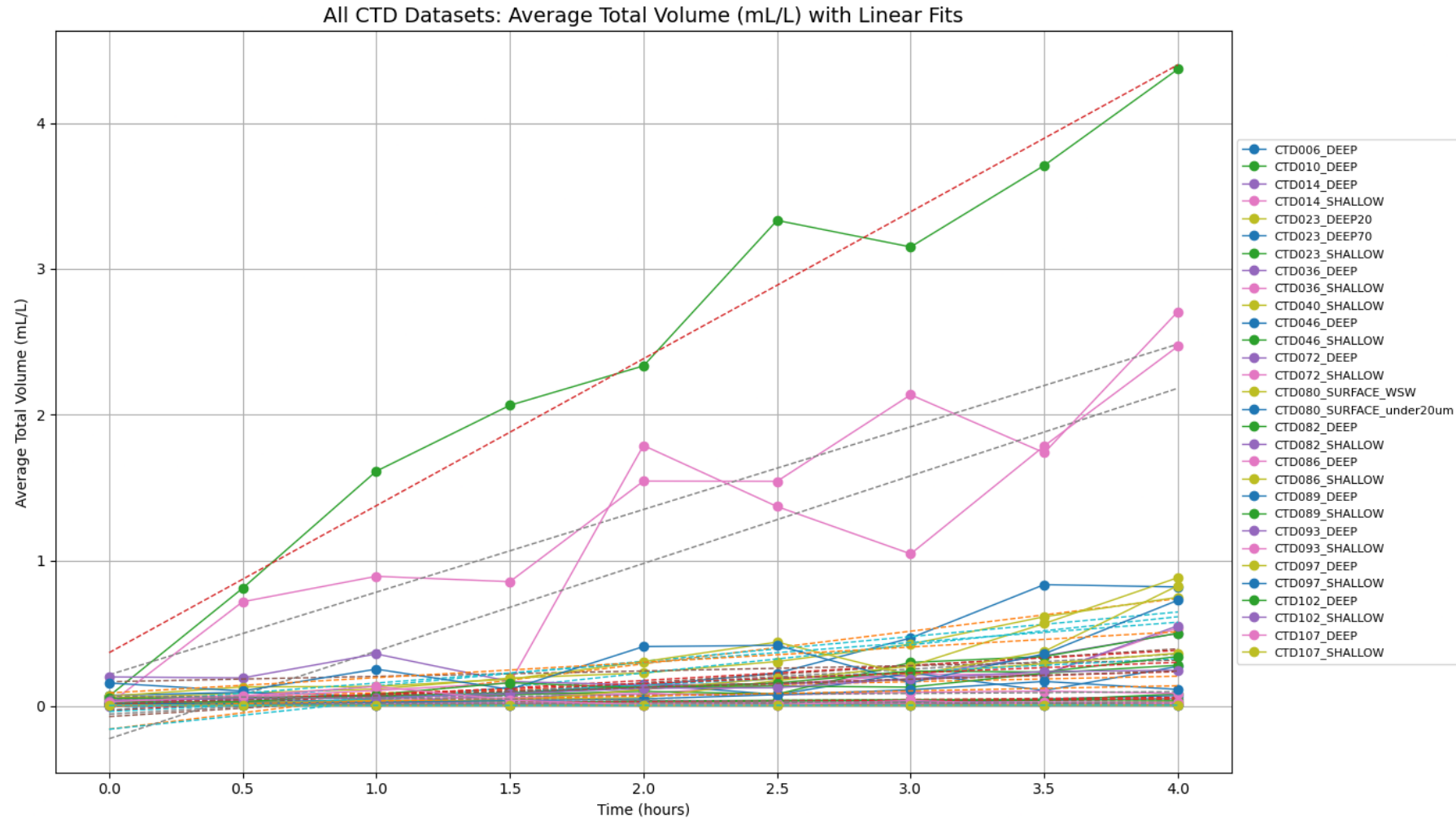
Roller Tank Processing Codes Final Version 5/12/26

Author: Ashely Kraekel, Dazhi Li

How this works

1. Settings
 - General setup, this version uses only 1 frame per timepoint, the actual used code averages data over 60 frames (30 before and 30 after the time point)
 - Has the volume of the liquid in the tank, use to change mL/L to L
 - Where the images are stored and where the results are saved
2. Calibration
 - The calibration value tells the code how many microns correspond to one pixel.

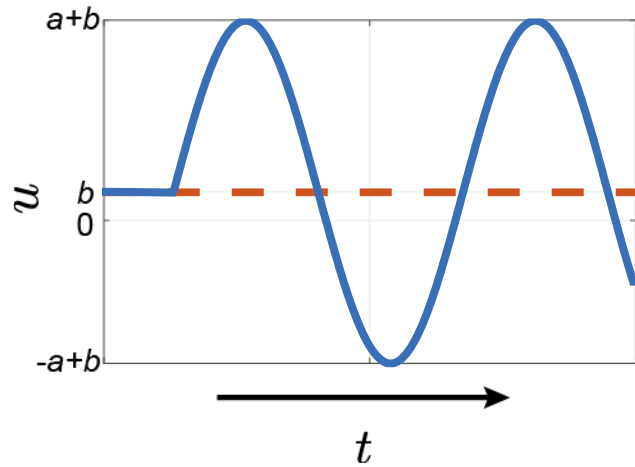
Below are example outputs from the roller tank for a 30 day cruise in June 2024, where CTD water was rolled for 4 hours



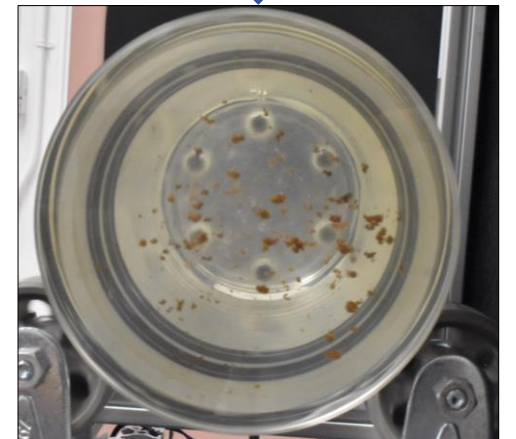
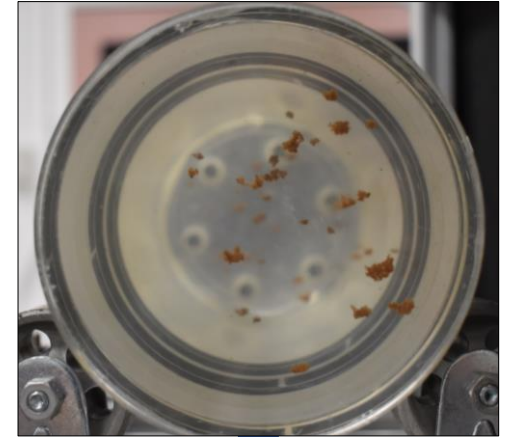
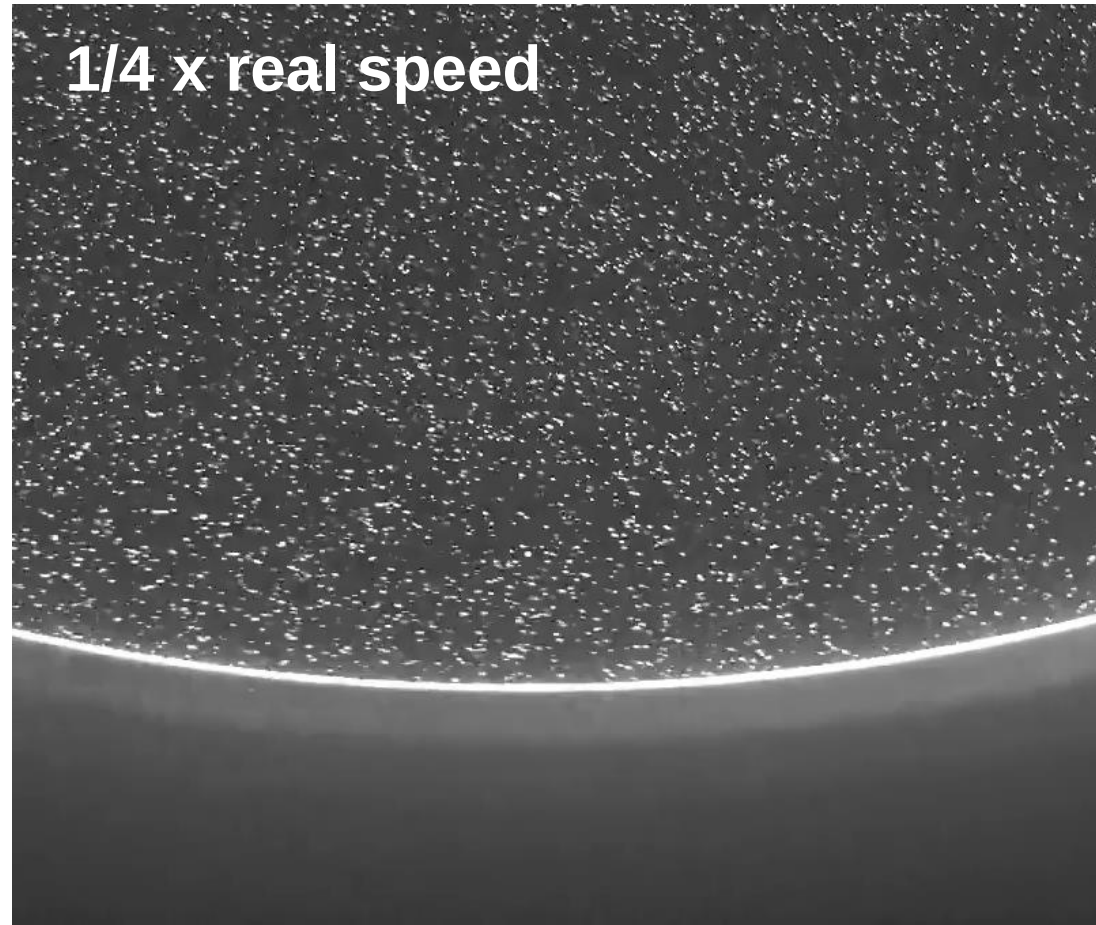
Accelerating the tank wall generates fluid shear



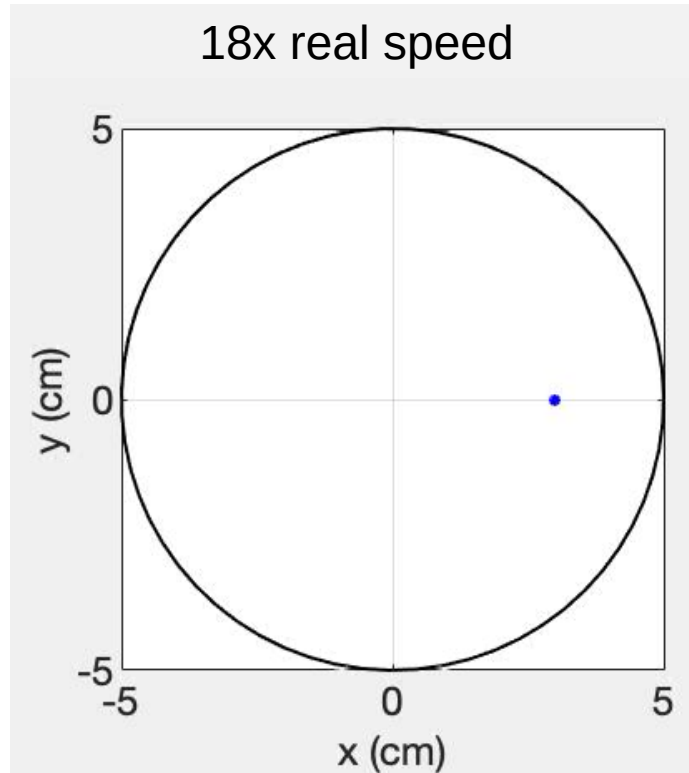
We superimposed a harmonic oscillation at the tank wall to sustain a periodic boundary layer with shear



Our modification for creating shear inside



The oscillation changes the 'stable regions' of particle motion in the tank. Some particles may hit the walls.

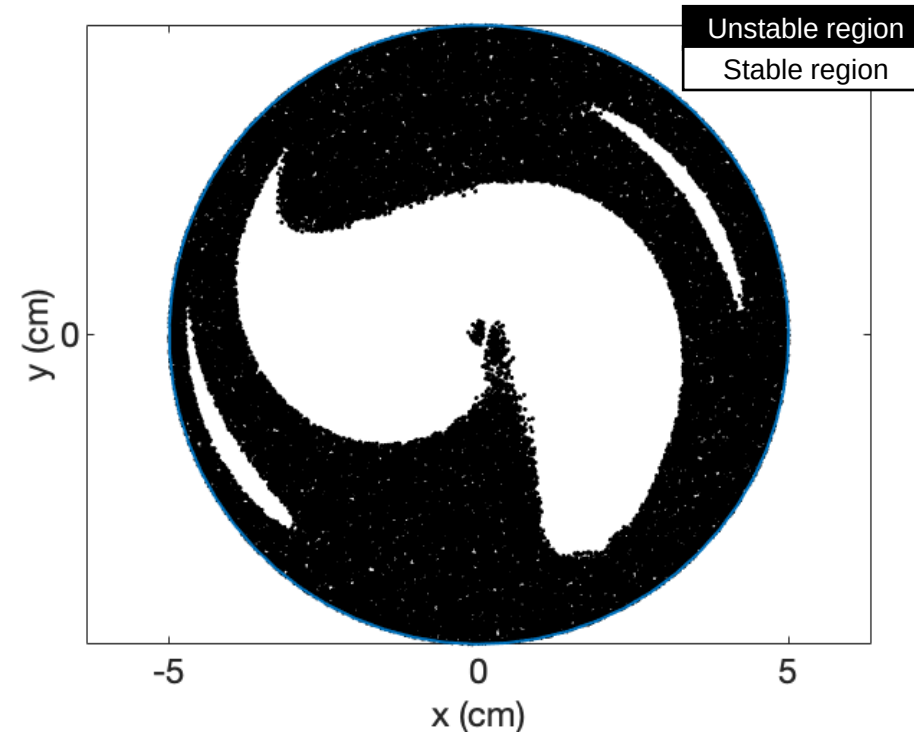


$$u_q(R, t) = 0.05 + 0.25 \sin(0.02t)$$

By seeding 100,000 points randomly,

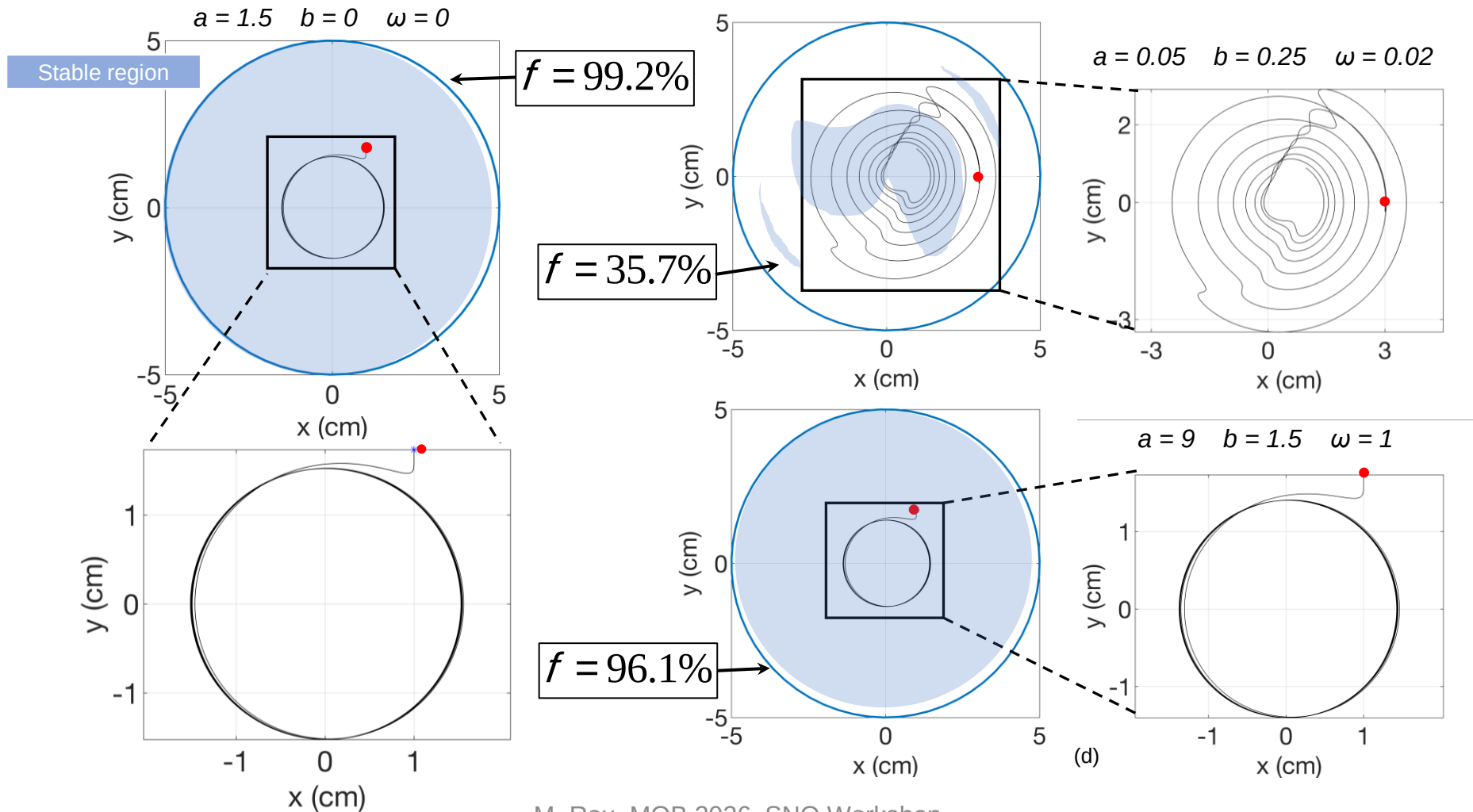
$$f = 35.7\%$$

for a 4.2 hour aggregation experiment.

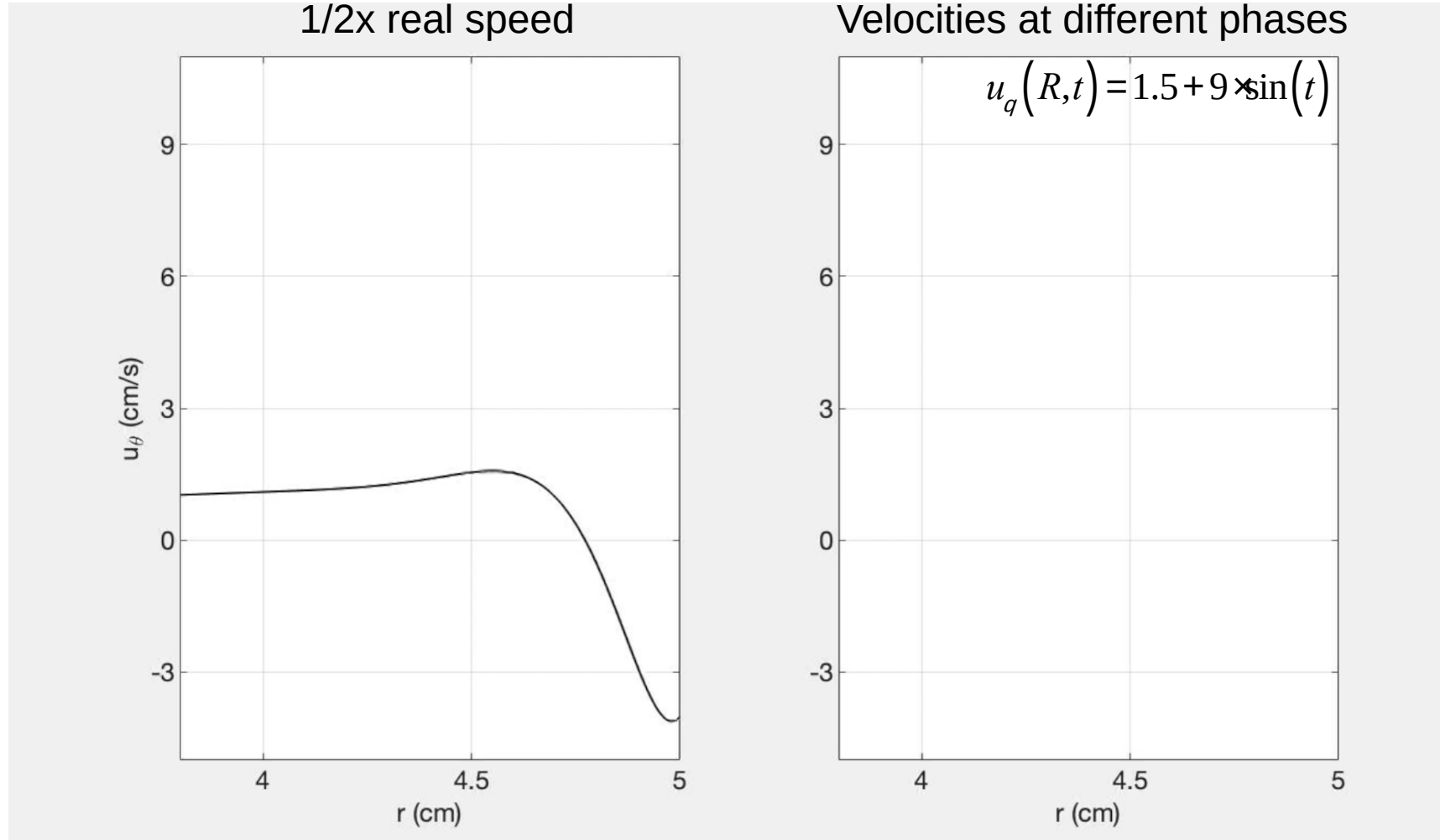


The stable regions can be tuned by altering the constant and oscillating speeds

$$u_q(R, t) = a + b \sin(\omega t)$$



The oscillation results in a large variation in velocity near the wall, which induces shear

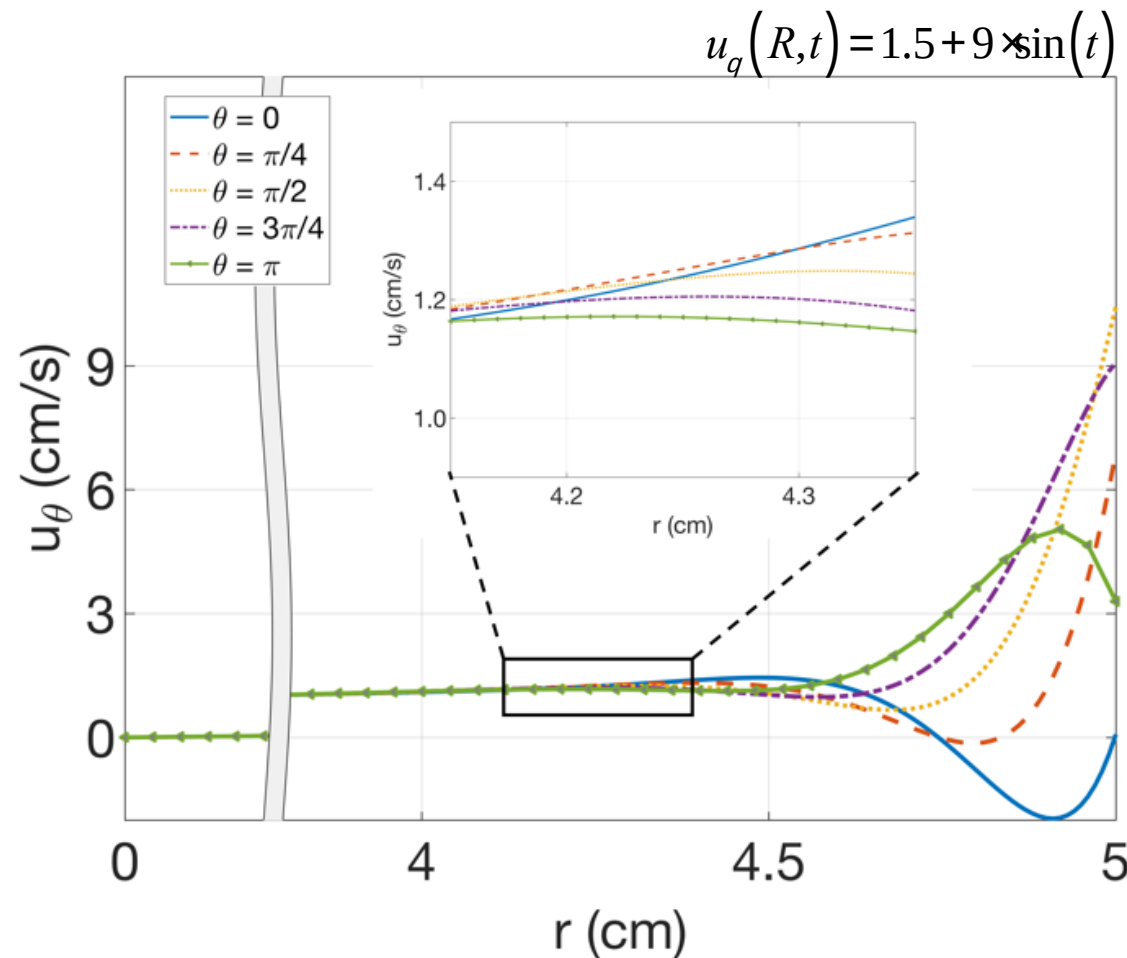


Shear is highest at the wall

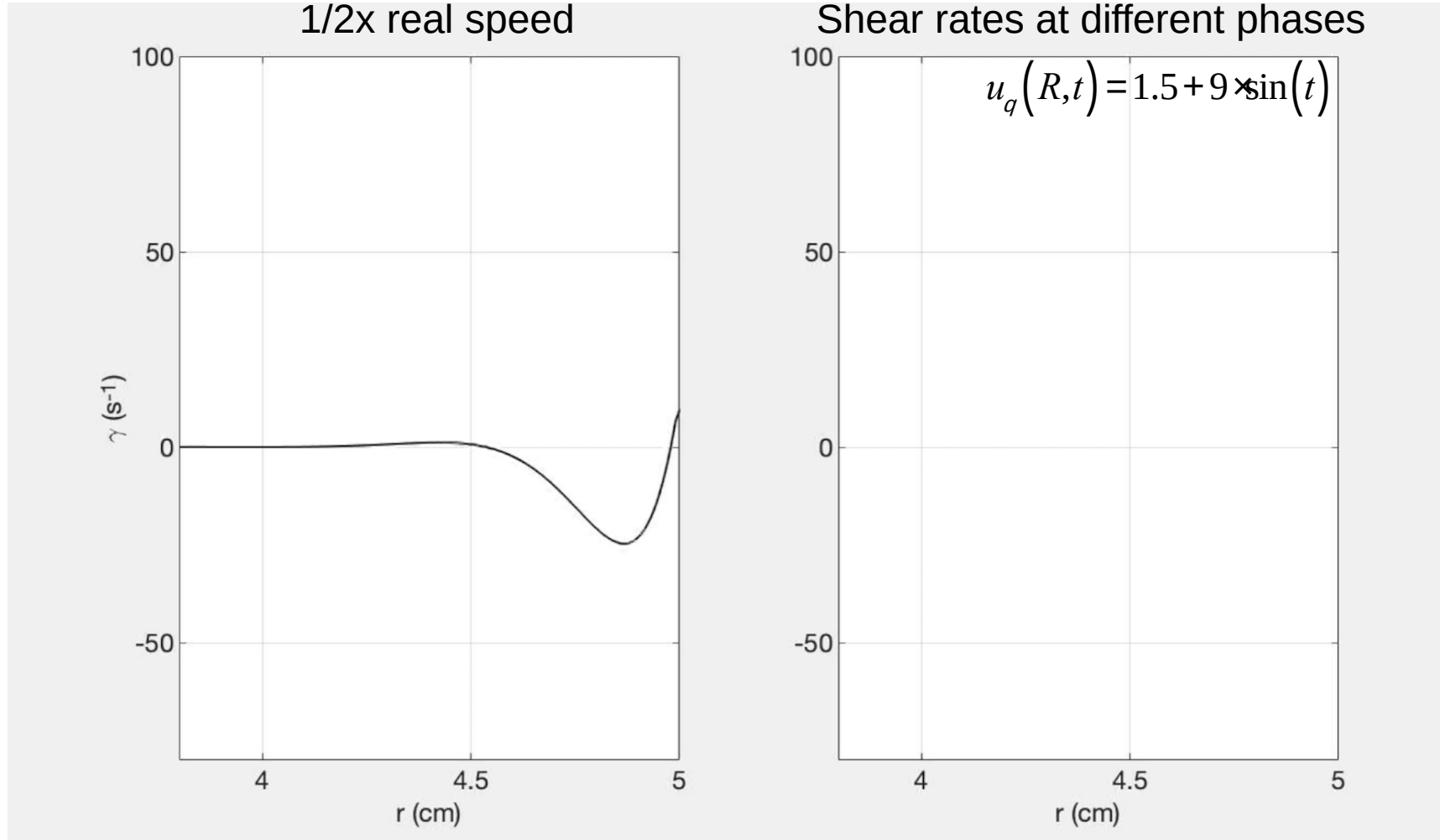
- The shear rate can be calculated as follows

$$g_{laminar} = r \frac{d(u_q / r)}{dr}$$

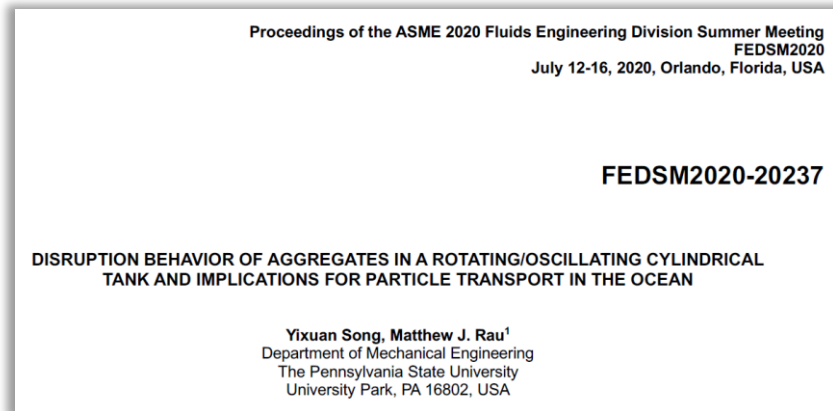
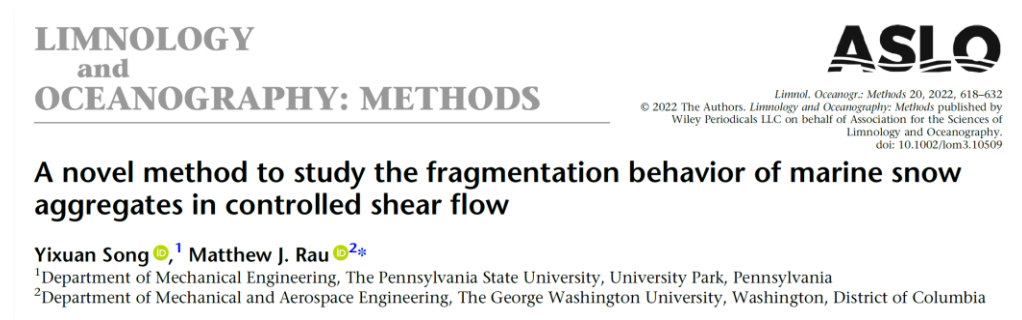
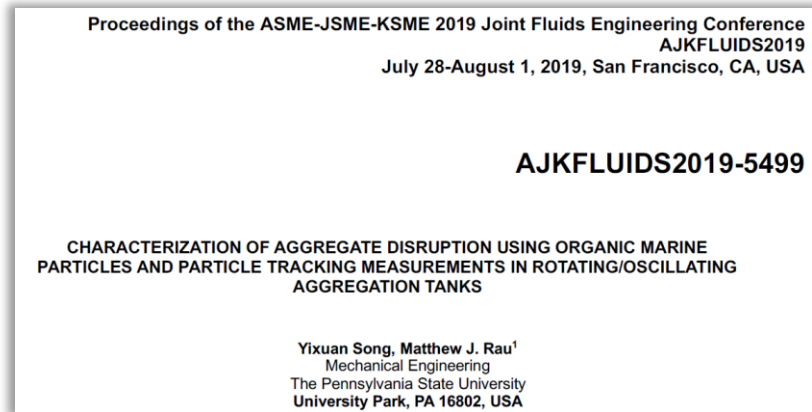
$$g_{turbulent} = \sqrt{\frac{e}{u}}$$



Shear is highest at the wall

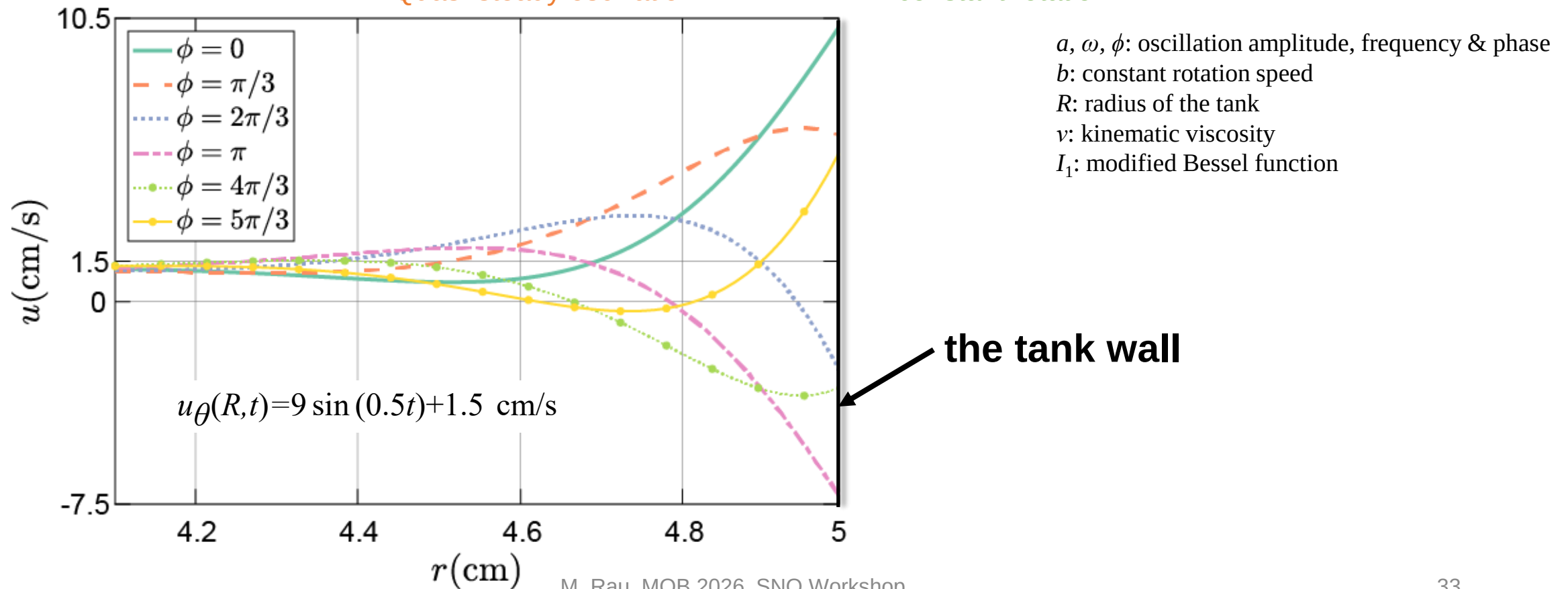


The tank is described here...

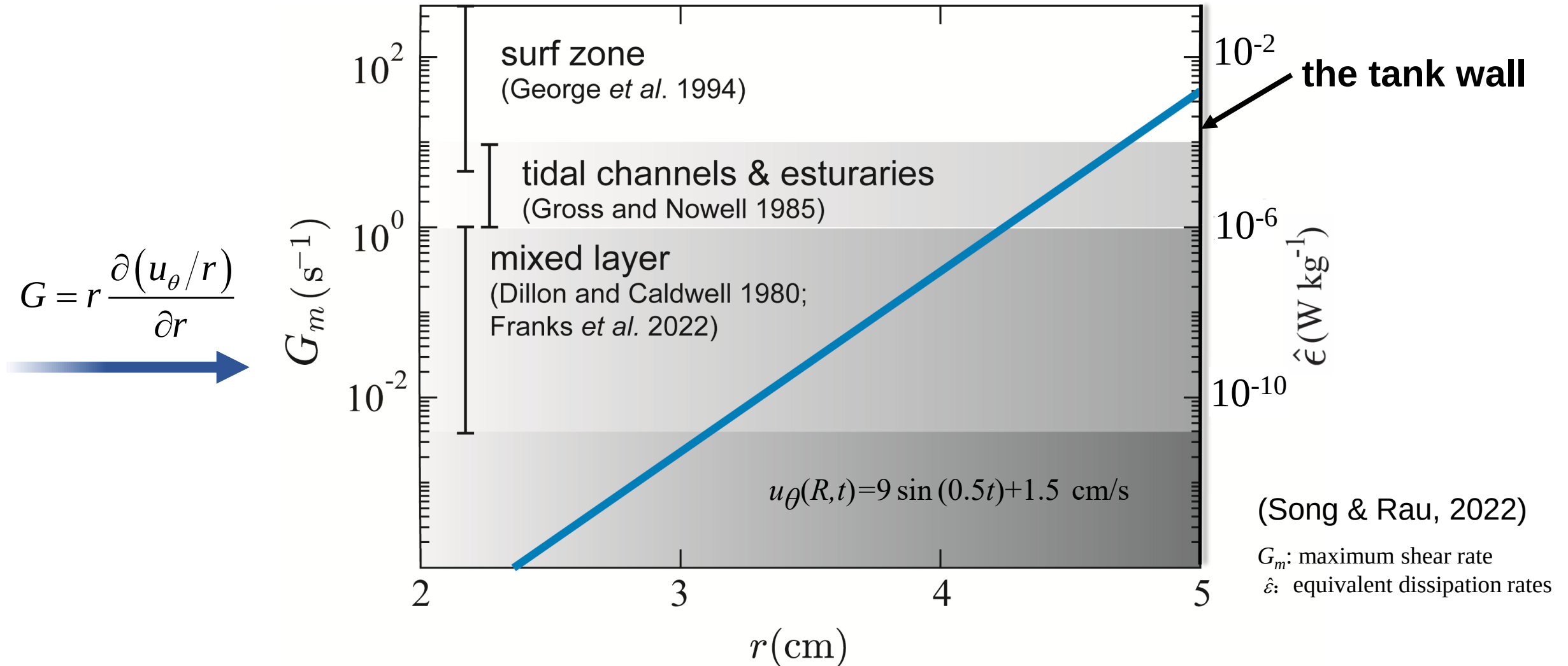


The flow is laminar and velocities and shear rates could be calculated analytically

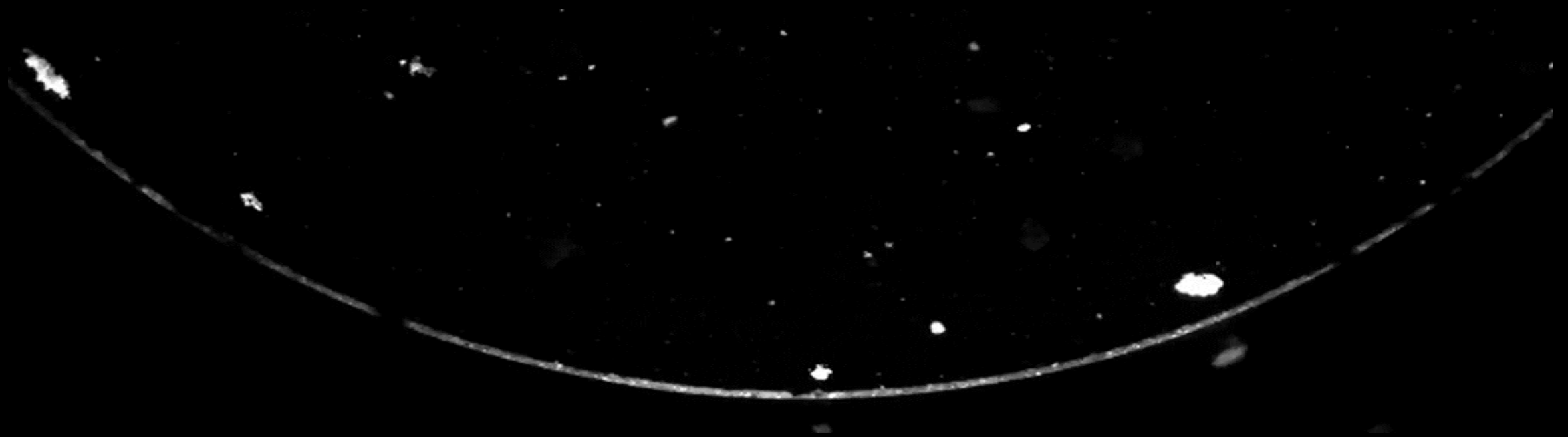
$$u_{\theta}(r,t) = \underbrace{\frac{a}{2} \left[\exp[i\omega t] \frac{I_1(r\sqrt{i\omega/\nu})}{I_1(R\sqrt{i\omega/\nu})} + \exp[-i\omega t] \frac{I_1(r\sqrt{-i\omega/\nu})}{I_1(R\sqrt{-i\omega/\nu})} \right]}_{\text{Quasi-steady oscillation}} + \underbrace{\frac{r}{R}b}_{\text{constant rotation}} \quad (\text{Song \& Rau, 2020})$$



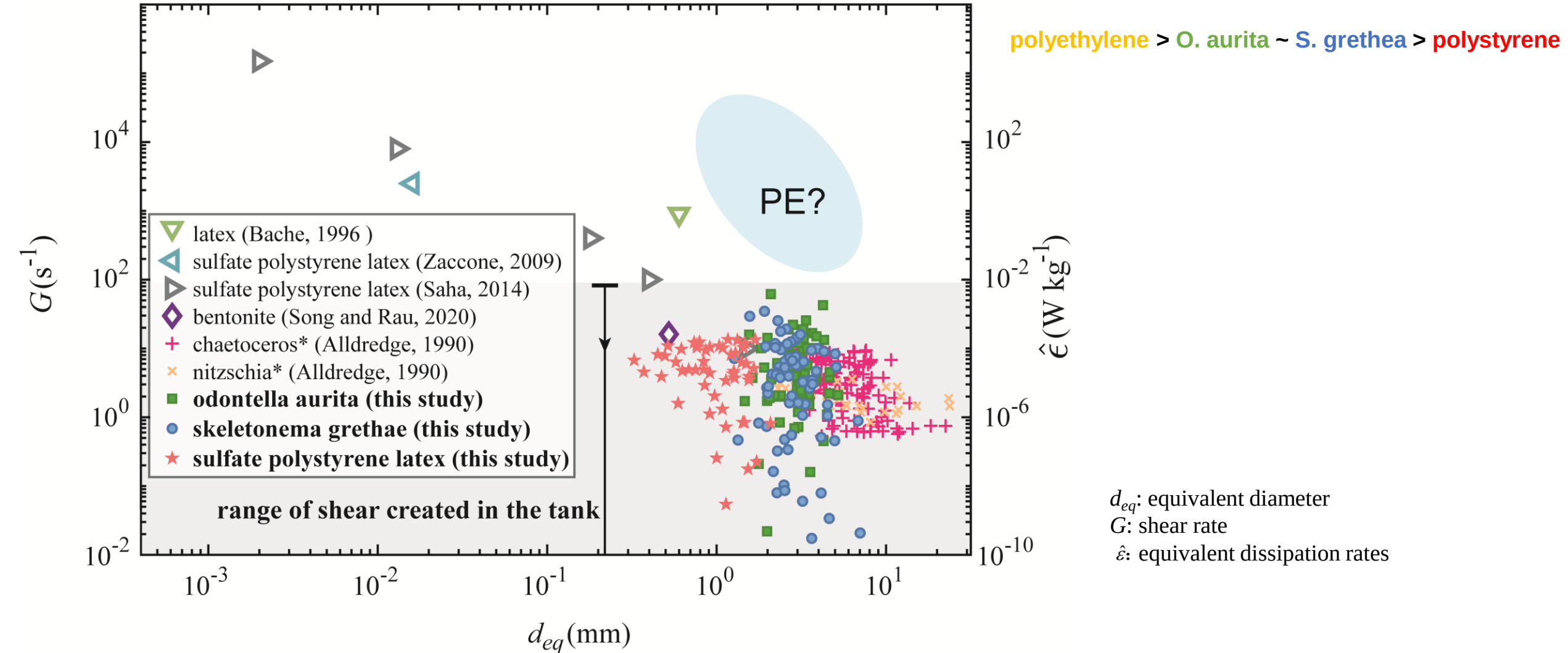
The tank provides shear rates similar to those in the ocean



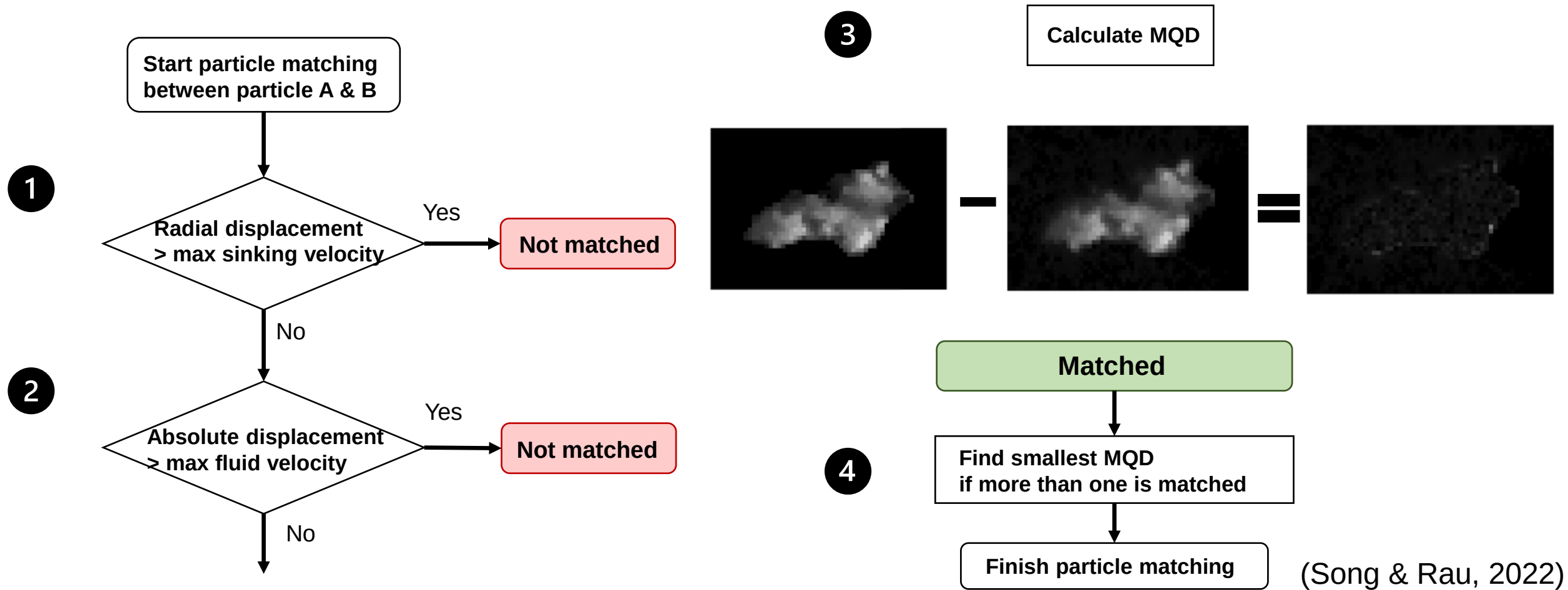
Aggregates in the higher shear regions can deform, elongate, and fragment



We found that phytoplankton aggregates can fragment at very low shear rates



But to determine fragmentation, we had to develop an algorithm to track aggregates and identify fragmentation events



But to determine fragmentation, we had to develop an algorithm to track aggregates and identify fragmentation events

	Frame #	<i>i</i>	<i>i</i> +1	<i>i</i> +2	<i>i</i> +3	<i>i</i> +4	<i>i</i> +5	<i>i</i> +6	
①	P	X	X	X					
	C1				X	X	X	X	Y
	C2				X	X	X	X	
②	P	X	X	X					
	C1					X	X	X	N
	C2				X	X	X	X	
③	P	X	X	X	X				
	C1		X			X	X	X	N
	C2		X			X	X	X	

	Frame #	<i>i</i>	<i>i</i> +1	<i>i</i> +2	<i>i</i> +3	<i>i</i> +4	<i>i</i> +5	<i>i</i> +6	
④	P	X		X					
	C1		X			X	X	X	N
	C2		X			X	X	X	

C1: potential child aggregate #1
C2: potential child aggregate #2
P: parent aggregate
Y: potential breakup event
N: not a breakup event

X	X	existence in a frame
		nonexistence in a frame

(Song & Rau, 2022)